

37 Flow of Falling Film
 129 4.4-1 Flow Near a Wallsudden
 537 Set in motion
 598

mixture $\rightarrow$ ^{species} velocities are different!

Diffusivity and the Mechanisms of Mass Transport

In Chapter 1 we began by stating Newton's law of viscosity, and in Chapter 8 we began with Fourier's law of heat conduction. In this chapter we present Fick's law of diffusion, which describes the movement of one species, say A , through a binary mixture of A and B because of the concentration gradient of A .

The movement of a chemical species from a region of high concentration to a region of low concentration can be observed with the naked eye by dropping a small crystal of potassium permanganate into a beaker of water. The KMnO_4 begins to dissolve in the water, and very near the crystal there is a dark purple concentrated solution of KMnO_4 . Because of the concentration gradient that is established, the KMnO_4 diffuses away from the crystal. The progress of the diffusion can then be followed by observing the growth of the purple region—dark purple where the KMnO_4 concentration is high and light purple where it is low.

Diffusion is more complicated than viscous flow or heat conduction because here for the first time we have to deal with mixtures. In a diffusing mixture the velocities of the individual species are different, and there are several useful ways of averaging the velocities of the species to get a local velocity for the mixture. It is necessary to choose such a local velocity before the

rates of diffusion can be defined. Hence it is appropriate that in §16.1 we discuss in some detail the definitions of concentrations, velocities, and fluxes. Although the discussion in §16.1 involves no new physical principles, a mastery of these definitions is essential for study of the later topics.

In §16.2 Fick's law of diffusion is stated, and the binary diffusivity $\mathcal{D}_{AB}$ is thereby defined. In §16.3 the temperature and pressure dependence of $\mathcal{D}_{AB}$ are discussed. In §16.4 and §16.5 the diffusivities of binary gas and liquid mixtures are considered from a molecular point of view.

The present lack of diffusivity data on most mixtures of interest makes it necessary to use estimated or extrapolated values of $\mathcal{D}_{AB}$ in most calculations. One should, of course, endeavor to use measured values, since they are usually (but not always) more reliable.

We have indicated that diffusion of A in a binary system of A and B occurs because of a concentration gradient of A . This phenomenon is sometimes called *ordinary diffusion* to distinguish it from *pressure diffusion* (motion of A resulting from a pressure gradient), *thermal diffusion* (motion of A resulting from a thermal gradient), and *forced diffusion* (motion of A because of unequal external forces on A and B). These additional effects, and also diffusion in multicomponent systems, are discussed briefly in Chapter 18.

Note: In this and the following chapters we adopt a fixed rule on notation. When considering *two-component systems*, we label the species A and B . When dealing with *multicomponent systems*, we label specific species 1, 2, 3, etc., or in general discussions we may use a dummy index such as i , j , or k to stand for the various species in turn. Formulas that are limited to binary systems are thus distinguished by the appearance of A and B as subscripts.

§16.1 DEFINITIONS OF CONCENTRATIONS, VELOCITIES, AND MASS FLUXES¹

In a multicomponent system the concentrations of the various species may be expressed in numerous ways. We limit our discussion to four: the *mass concentration*, ρ_i , which is the mass of species i per unit volume of solution; the *molar concentration*, $c_i = \rho_i/M_i$, which is the number of moles of species i per unit volume of solution; the *mass fraction*, $\omega_i = \rho_i/\rho$, which is the mass concentration of species i divided by the total mass density of the solution; and the *mole fraction*, $x_i = c_i/c$, which is the molar concentration of species i divided by the total molar density of the solution. (By the word "solution" we mean a one-phase gaseous, liquid, or solid mixture.) In Table 16.1-1 a summary is given of these concentration units and their interrelation for binary systems.

¹ R. B. Bird, Theory of Diffusion, in *Advances in Chemical Engineering*, Vol. I, Academic Press, New York (1956), pp. 170, *et seq.*

In a diffusing mixture the various chemical species are moving at different velocities. Let v_i denote the velocity of the i th species with respect to stationary coordinate axes.² Then, for a mixture of n species, the local *mass average velocity* v is defined as

$$v = \frac{\sum_{i=1}^n \rho_i v_i}{\sum_{i=1}^n \rho_i} \quad (16.1-1)$$

Note that ρv is the local rate at which mass passes through a unit cross section placed perpendicular to the velocity v . This is the local velocity one would measure by means of a pitot tube and corresponds to v as used in the preceding chapters for pure fluids. Similarly, one may define a local *molar average velocity* v^* as

$$v^* = \frac{\sum_{i=1}^n c_i v_i}{\sum_{i=1}^n c_i} \quad (16.1-2)$$

Note that cv^* is the local rate at which moles pass through a unit cross section placed perpendicular to the velocity v^* . We shall find extensive use for both of these average velocities. Still other average velocities are sometimes used, such as the *volume average velocity* v^∇ . (See Problem 16.K.)

In flow systems one is frequently interested in the velocity of a given species with respect to v or v^* rather than with respect to stationary coordinates. This leads to the definition of the "diffusion velocities":

$$v_i - v = \text{diffusion velocity of } i \text{ with respect to } v \quad (16.1-3)$$

$$v_i - v^* = \text{diffusion velocity of } i \text{ with respect to } v^* \quad (16.1-4)$$

These diffusion velocities indicate the motion of component i relative to the local motion of the fluid stream.

The notation for velocities in binary systems is summarized in Table 16.1-2, together with some useful relations among the various velocities.

It is important that the reader thoroughly understand the meaning of the various kinds of velocities. Therefore, let us examine a specific system and represent its behavior pictorially. In Fig. 16.1-1 we consider what happens when liquid A evaporates and diffuses upward through a long tube initially filled with vapor B . As A evaporates, it pushes gas B upward. However, there is not a clear dividing line between pure A vapor and pure B vapor, as suggested in Fig. 16.1-1a; instead, the upward displacement of B is

² By "velocity" we do not mean the velocity of an individual molecule of species i ; rather we mean the sum of the velocities of the molecules of species i within a small volume element divided by the number of such molecules.

TABLE 16.1-1
NOTATION FOR CONCENTRATIONS IN BINARY SYSTEMS

Basic definitions	$\rho = \rho_A + \rho_B = \text{mass density of solution (g/cm}^3\text{)}$ (A)
	$\rho_A = c_A M_A = \text{mass concentration of } A \text{ (g of } A/\text{cm}^3 \text{ of solution)}$ (B)
	$\omega_A = \frac{\rho_A}{\rho} = \text{mass fraction of } A$ (C)
	$c = c_A + c_B = \text{molar density of solution (g-moles/cm}^3\text{)}$ (D)
	$c_A = \frac{\rho_A}{M_A} = \text{molar concentration of } A \text{ (g-moles of } A/\text{cm}^3 \text{ of solution)}$ (E)
	$x_A = \frac{c_A}{c} = \text{mole fraction of } A$ (F)
	$M = \frac{\rho}{c} = \text{number-mean molecular weight of mixture}$ (G)
Additional relations, for reference only	$x_A + x_B = 1$ (H)
	$x_A M_A + x_B M_B = M$ (J)
	$x_A = \frac{\frac{\omega_A}{M_A}}{\frac{\omega_A}{M_A} + \frac{\omega_B}{M_B}}$ (L)
	$\omega_A + \omega_B = 1$ (I)
	$\frac{\omega_A}{M_A} + \frac{\omega_B}{M_B} = \frac{1}{M}$ (K)
	$\omega_A = \frac{x_A M_A}{x_A M_A + x_B M_B}$ (M)
	$dx_A = \frac{d\omega_A}{M_A M_B \left(\frac{\omega_A}{M_A} + \frac{\omega_B}{M_B} \right)^2}$ (N)
	$d\omega_A = \frac{M_A M_B dx_A}{(x_A M_A + x_B M_B)^2}$ (O)

TABLE 16.1-2
NOTATION FOR VELOCITIES IN BINARY SYSTEMS

Basic definitions	$v_A = \text{velocity of species } A \text{ relative to stationary coordinates}$ (A)
	$v_A - v = \text{diffusion velocity of species } A \text{ relative to } v$ (B)
	$v_A - v^* = \text{diffusion velocity of species } A \text{ relative to } v^*$ (C)
	$v = \text{mass average velocity} = (1/\rho)(\rho_A v_A + \rho_B v_B) = \omega_A v_A + \omega_B v_B$ (D)
	$v^* = \text{molar average velocity} = (1/c)(c_A v_A + c_B v_B) = x_A v_A + x_B v_B$ (E)
Additional relations	$v - v^* = \omega_A(v_A - v^*) + \omega_B(v_B - v^*)$ (F)
	$v^* - v = x_A(v_A - v) + x_B(v_B - v)$ (G)

TABLE 16.1-3
MASS AND MOLAR FLUXES IN BINARY SYSTEMS

Quantity	With Respect to Stationary Axes	With Respect to v	With Respect to v^*
Velocity of species A (cm sec ⁻¹)	v_A (A)	$v_A - v$ (B)	$v_A - v^*$ (C)
Mass flux of species A (g cm ⁻² sec ⁻¹)	$n_A = \rho_A v_A$ (D)	$j_A = \rho_A(v_A - v)$ (E)	$j_A^* = \rho_A(v_A - v^*)$ (F)
Molar flux of species A (g-moles cm ⁻² sec ⁻¹)	$N_A = c_A v_A$ (G)	$J_A = c_A(v_A - v)$ (H)	$J_A^* = c_A(v_A - v^*)$ (I)
Sum of mass fluxes (g cm ⁻² sec ⁻¹)	$n_A + n_B = \rho v$ (J)	$j_A + j_B = 0$ (K)	$j_A^* + j_B^* = \rho(v - v^*)$ (L)
Sum of molar fluxes (g-moles cm ⁻² sec ⁻¹)	$N_A + N_B = cv^*$ (M)	$J_A + J_B = c(v^* - v)$ (N)	$J_A^* + J_B^* = 0$ (O)
Fluxes in terms of n_A and n_B	$N_A = \frac{n_A}{M_A}$ (P)	$j_A = n_A - \omega_A(n_A + n_B)$ (Q)	$j_A^* = n_A - x_A\left(n_A + \frac{M_A}{M_B}n_B\right)$ (R)
Fluxes in terms of N_A and N_B	$n_A = N_A M_A$ (S)	$J_A = N_A - \omega_A\left(N_A + \frac{M_B}{M_A}N_B\right)$ (T)	$J_A^* = N_A - x_A(N_A + N_B)$ (U)
Fluxes in terms of j_A and v	$n_A = j_A + \rho_A v$ (V)	$J_A = \frac{j_A}{M_A}$ (W)	$j_A^* = \frac{M}{M_B}j_A$ (X)
Fluxes in terms of J_A^* and v^*	$N_A = J_A^* + c_A v^*$ (Y)	$J_A = \frac{M_B}{M}J_A^*$ (Z)	$j_A^* = J_A^* M_A$ (AA)

accompanied by an intermingling of the two vapors, as shown in Fig. 16.1-1b. Thus at any point in the column A is moving upward more rapidly than the over-all average motion and B is moving less rapidly, because of diffusion. In Fig. 16.1-1b the various velocity vectors are shown for the situation that $x_A = \frac{1}{6}$, $v^* = 12$, $v_A - v^* = 3$, and $M_A = 5M_B$. The reader should verify the other velocity vectors.

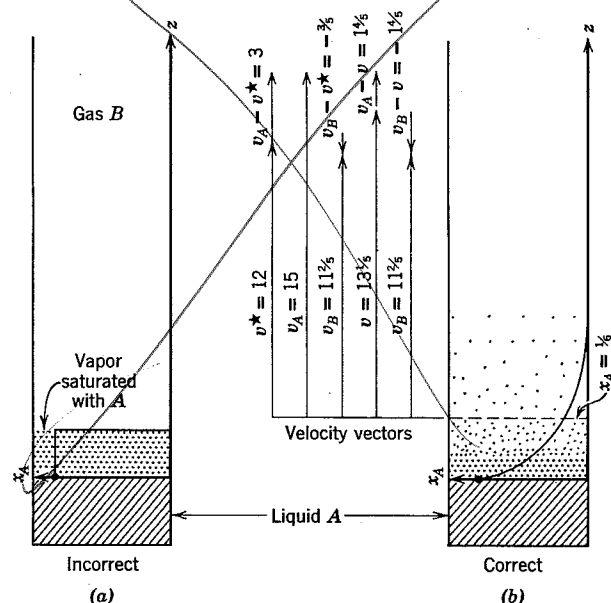


Fig. 16.1-1. Concentration distribution at a specific time in a diffusing system.

Now that concentrations and velocities have been discussed, we are in a position to define mass and molar fluxes. The mass (or molar) flux of species i is a vector quantity denoting the mass (or moles) of species i that passes through a unit area per unit time. The motion may be referred to stationary coordinates, to the local mass average velocity v , or to the local molar average velocity v^* . Thus the mass and molar fluxes *relative to stationary coordinates* are

$$n_i = \rho_i v_i \quad \text{mass} \quad (16.1-5)$$

$$N_i = c_i v_i \quad \text{molar} \quad (16.1-6)$$

The mass and molar fluxes *relative to the mass-average velocity* v are

$$j_i = \rho_i (v_i - v) \quad \text{mass} \quad (16.1-7)$$

$$J_i = c_i (v_i - v) \quad \text{molar} \quad (16.1-8)$$

and the mass and molar fluxes *relative to the molar-average velocity* v^* are

$$j_i^* = \rho_i (v_i - v^*) \quad \text{mass} \quad (16.1-9)$$

$$J_i^* = c_i (v_i - v^*) \quad \text{molar} \quad (16.1-10)$$

The notation for fluxes in binary systems is summarized in Table 16.1-3, together with some relations among the various fluxes. It should be emphasized that the definition of a mass flux is not complete until both *units* and *reference frame* have been given.

Mathematically speaking, any *one* of the foregoing six flux notations is adequate for all diffusion problems, but each has certain advantages and all have appeared in the literature. The flux N_i (and to a lesser extent n_i) has been used in engineering because in process calculations it is usually desirable to refer to a coordinate system fixed in the equipment. The fluxes j_i and J_i^* are the usual measures of diffusion rates and are useful in formulating the equations of change for multicomponent systems. The fluxes J_i and j_i^* are seldom used but are included here for the sake of completeness. We use N_i in most subsequent discussions.

Example 16.1-1. Relations Among the Molar Fluxes

- How are the fluxes J_i^* and N_i related in an n -component system?
- Show that the sum of the fluxes J_i^* is zero.

Solution. *a.* First we combine the definitions of v^* and J_i^* (Eqs. 16.1-2 and 10) to get

$$\begin{aligned} J_i^* &= c_i (v_i - v^*) \\ &= c_i v_i - \frac{c_i}{c} \sum_{j=1}^n c_j v_j \end{aligned} \quad (16.1-11)$$

Application of the definitions of N_i and x_i then gives

$$J_i^* = N_i - x_i \sum_{j=1}^n N_j \quad (16.1-12)$$

From this result it can be seen that the molar diffusion flux J_i^* is simply the difference between the molar flux N_i and the rate of bodily transport (bulk flow) of species i due to the local molar flux of the mixture. Equation 16.1-10 may be similarly interpreted.

- Summation of Eq. 16.1-12 from $i = 1$ to $i = n$ gives

$$\sum_{i=1}^n J_i^* = 0 \quad (16.1-13)$$

which shows that the sum of the *molar* diffusion fluxes relative to the *molar* average velocity is zero in any mixture. Thus in a binary mixture the diffusion fluxes J_A^* and J_B^* are of equal magnitude and are oppositely directed:

$$J_A^* = -J_B^* \quad (16.1-14)$$

An application of this result is given in Problem 16.I.

§16.2 FICK'S LAW OF DIFFUSION

In Eq. 1.1-2 the viscosity μ is defined as the proportionality factor between momentum flux and velocity gradient (Newton's law of viscosity). In Eq. 8.1-6 the thermal conductivity k is defined as the proportionality factor between heat flux and temperature gradient (Fourier's law of heat conduction).

TABLE 16.2-1

EQUIVALENT FORMS OF FICK'S FIRST LAW OF BINARY DIFFUSION

Flux	Gradient	Form of Fick's First Law
n_A	$\nabla \omega_A$	$n_A - \omega_A(n_A + n_B) = -\rho \mathcal{D}_{AB} \nabla \omega_A$ (A)
N_A	∇x_A	$N_A - x_A(N_A + N_B) = -c \mathcal{D}_{AB} \nabla x_A$ (B)
j_A	$\nabla \omega_A$	$j_A = -\rho \mathcal{D}_{AB} \nabla \omega_A$ (C)
J_A	∇x_A	$J_A = -c \mathcal{D}_{AB} \nabla x_A$ (D)
j_A^*	∇x_A	$j_A^* = -\left(\frac{c^2}{\rho}\right) M_A M_B \mathcal{D}_{AB} \nabla x_A$ (E)
J_A^*	$\nabla \omega_A$	$J_A^* = -\left(\frac{\rho^2}{c M_A M_B}\right) \mathcal{D}_{AB} \nabla \omega_A$ (F)
$c(v_A - v_B)$	∇x_A	$c(v_A - v_B) = -\frac{c \mathcal{D}_{AB}}{x_A x_B} \nabla x_A$ (G)

Now we define the mass diffusivity $\mathcal{D}_{AB} = \mathcal{D}_{BA}$ in a binary system in an analogous fashion:

$$J_A^* = -c \mathcal{D}_{AB} \nabla x_A \quad (16.2-1)$$

This is *Fick's first law of diffusion*,¹ written in terms of the molar diffusion flux J_A^* . This equation states that species A diffuses (moves relative to the mixture) in the direction of decreasing mole fraction of A , just as heat flows by conduction in the direction of decreasing temperature.

A number of other mathematically equivalent statements of Fick's first law have appeared in the literature, and some of them are summarized in Table 16.2-1 for reference only. The diffusivity $\mathcal{D}_{AB}$ is *identical* in all these equations. Of special importance in the following chapters is the form of Fick's first law in terms of N_A , the molar flux relative to stationary coordinates:

$$N_A = x_A(N_A + N_B) - c \mathcal{D}_{AB} \nabla x_A \quad (16.2-2)$$

This equation shows that the diffusion flux N_A relative to stationary coordinates is the *resultant* of two *vector* quantities: the vector $x_A(N_A + N_B)$,

¹ Temperature gradients, pressure gradients, and external forces also contribute to the diffusion flux, although their effects are usually minor. More complete expressions for the diffusion flux are given in §18.4 [A. Fick, *Ann. der Physik*, **94**, 59-86 (1855).]

which is the molar flux of A resulting from the bulk motion of the fluid, and the vector $J_A^* = -c \mathcal{D}_{AB} \nabla x_A$, which is the molar flux of A resulting from the diffusion superimposed on the bulk flow. Thus in Fig. 16.1-1b the bulk flow and diffusion terms in Eq. 16.2-2 are in the same direction for species A (because A is diffusing with the current) and are opposed for species B (because B is diffusing against the current).

The units of the mass diffusivity $\mathcal{D}_{AB}$ are $\text{cm}^2 \text{sec}^{-1}$ or $\text{ft}^2 \text{hr}^{-1}$. Note that the kinematic viscosity ν and the thermal diffusivity α also have the same units. The way in which these three quantities are analogous can be seen from the following equations for the fluxes of mass, momentum, and energy in one-dimensional systems:

$$j_{Av} = -\mathcal{D}_{AB} \frac{d}{dy} (\rho_A) \quad (\text{Fick's law}^1 \text{ for constant } \rho) \quad (16.2-3)$$

$$\tau_{yx} = -\nu \frac{d}{dy} (\rho v_x) \quad (\text{Newton's law for constant } \rho) \quad (16.2-4)$$

$$q_y = -\alpha \frac{d}{dy} (\rho \hat{C}_p T) \quad (\text{Fourier's law for constant } \rho \hat{C}_p) \quad (16.2-5)$$

These equations state, respectively, that (a) mass transport occurs because of a gradient in mass concentration, (b) momentum transport occurs because of a gradient in momentum concentration, and (c) energy transport occurs because of a gradient in energy concentration. These analogies do not apply in two- and three-dimensional problems, however, because τ is a tensor quantity with nine components, whereas j_A and q are vectors with three components.

In Tables 16.2-2, 3, and 4 some values of $\mathcal{D}_{AB}$ are given for a few gas,

TABLE 16.2-2
EXPERIMENTAL DIFFUSIVITIES OF SOME DILUTE GAS PAIRS^a

Gas Pair	Temperature (° K)	$\mathcal{D}_{AB}$ ($\text{cm}^2 \text{sec}^{-1}$)
CO ₂ -N ₂ O	273.2	0.096
CO ₂ -CO	273.2	0.139
CO ₂ -N ₂	273.2	0.144
	288.2	0.158
	298.2	0.165
Ar-O ₂	293.2	0.20
H ₂ -SF ₆	298.2	0.420
H ₂ -CH ₄	298.2	0.726

^a This table is abstracted from J. O. Hirschfelder, C. F. Curtiss, and R. B. Bird, *Molecular Theory of Gases and Liquids*, Wiley, New York (1954), p. 579. The values given are for 1 atm pressure.

TABLE 16.2-3
 EXPERIMENTAL DIFFUSIVITIES IN THE LIQUID STATE^a

A	B	T (°C)	x_A	$\mathcal{D}_{AB} \times 10^5$ (cm ² sec ⁻¹)
Chlorobenzene	Bromobenzene	10.01	0.0332	1.007
			0.2642	1.069
			0.5122	1.146
			0.7617	1.226
			0.9652	1.291
		39.97	0.0332	1.584
			0.2642	1.691
			0.5122	1.806
			0.7617	1.902
			0.9652	1.996
Ethanol	Water	25	0.05	1.13
			0.275	0.41
			0.50	0.90
			0.70	1.40
			0.95	2.20
Water	n-Butanol	30	0.131	1.24
			0.222	0.920
			0.358	0.560
			0.454	0.437
			0.524	0.267

^a This table is abstracted from a review article by P. A. Johnson and A. L. Babb, Liquid Diffusion in Non-Electrolytes, *Chem. Revs.*, **56**, 387-453 (1956); in this article a summary of experimental diffusion coefficients for liquid systems is given, as well as a survey of methods of measurement. Another excellent review article is that of L. J. Gosting, Measurement and Interpretation of Diffusion Coefficients of Proteins, *Advances in Protein Chemistry*, Vol. XI, Academic Press, New York (1956).

liquid, and solid systems. Diffusivities of gases at low density are almost composition independent, increase with the temperature, and vary inversely with pressure. Liquid and solid diffusivities are strongly concentration-dependent and generally increase with temperature. In the next three sections we summarize the available means for estimating diffusivities.

§16.3 TEMPERATURE AND PRESSURE DEPENDENCE OF MASS DIFFUSIVITY

The mass diffusivity $\mathcal{D}_{AB}$ for a binary system is a function of temperature, pressure, and composition, whereas the viscosity μ and thermal conductivity

 TABLE 16.2-4
 EXPERIMENTAL DIFFUSIVITIES IN THE SOLID STATE^a

System	T (°C)	Diffusivity, $\mathcal{D}_{AB}$ (cm ² sec ⁻¹)
He in SiO ₂	20	$2.4 - 5.5 \times 10^{-10}$
	20	4.5×10^{-11}
	500	2×10^{-8}
H ₂ in SiO ₂	500	$0.6 - 2.1 \times 10^{-8}$
	85	1.16×10^{-8}
	165	10.5×10^{-8}
Bi in Pb	20	1.1×10^{-16}
Hg in Pb	20	2.5×10^{-15}
Sb in Ag	20	3.5×10^{-21}
Al in Cu	20	1.3×10^{-30}
Cd in Cu	20	2.7×10^{-15}

^a Values taken from R. M. Barrer, *Diffusion in and through Solids*, Macmillan, New York (1941), pp. 141, 222, and 275.

k for a pure fluid are functions only of temperature and pressure. The data available on $\mathcal{D}_{AB}$ for most binary mixtures are, moreover, quite limited in range and accuracy. The available correlations of $\mathcal{D}_{AB}$ are of limited scope and are based more on theory than on experiment.

For binary gas mixtures at low pressure, $\mathcal{D}_{AB}$ is inversely proportional to the pressure, increases with increasing temperature, and is almost independent of composition for a given gas-pair. The following equation for estimation of $\mathcal{D}_{AB}$ at low pressures has been developed¹ from a combination of kinetic theory and corresponding-states arguments:

$$\frac{p\mathcal{D}_{AB}}{(p_{cA}p_{cB})^{1/2}(T_{cA}T_{cB})^{1/2}} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{1/2} = a \left(\frac{T}{\sqrt{T_{cA}T_{cB}}} \right)^b \quad (16.3-1)$$

in which $\mathcal{D}_{AB}$ [=] cm² sec⁻¹, p [=] atm, and T [=] °K. Analysis of experimental data gave the following values of the constants a and b :

For nonpolar gas-pairs:

$$a = 2.745 \times 10^{-4}$$

$$b = 1.823$$

For H₂O with a nonpolar gas:

$$a = 3.640 \times 10^{-4}$$

$$b = 2.334$$

¹ J. C. Slattery and R. B. Bird, *A.I.Ch.E. Journal*, **4**, 137-142 (1958).

Equation 16.3-1 agrees with experimental data at atmospheric pressure within about 8 per cent.^{1,2} If gases *A* and *B* are nonpolar and their Lennard-Jones parameters are available, the kinetic theory method given in §16.4 will usually give better accuracy.

At high pressures $\mathcal{D}_{AB}$ no longer varies inversely with the pressure. Very little is known about this pressure dependence, except in the limiting case of self-diffusion (interdiffusion of identical molecules), which can be closely

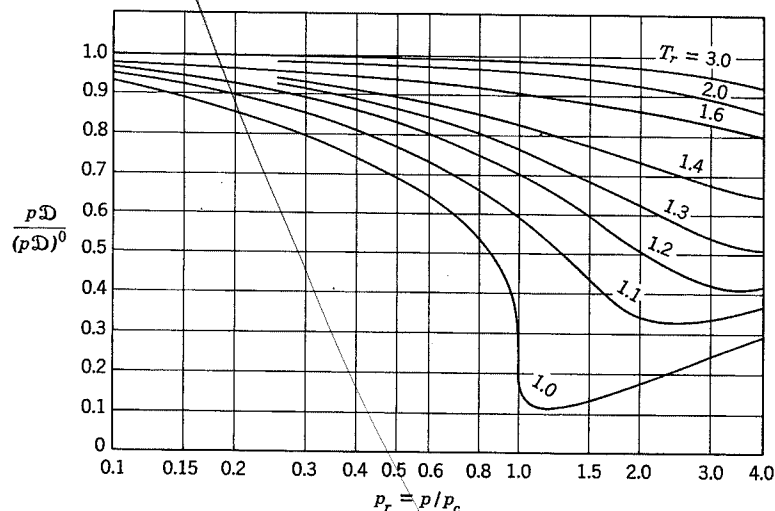


Fig. 16.3-1. Generalized chart for self-diffusivities of dense gases [J. C. Slattery, M.S. thesis, University of Wisconsin (1955)]; this chart is based on very few data.

approached experimentally by using isotopic tracers. The pressure dependence of the self-diffusivity, $\mathcal{D}_{AA}$, may be estimated from Fig. 16.3-1; this plot is based on the Enskog kinetic theory of dense gases and on some fragmentary data.¹ The ordinate of the plot is $p\mathcal{D}_{AA}/(p\mathcal{D}_{AA})^0$, the ratio of the pressure-diffusivity product at pressure *p* and temperature *T* to the pressure-diffusivity product at the same temperature but at low pressure. This ratio is plotted as a function of the reduced temperature $T_r = T/T_c$ and the reduced pressure $p_r = p/p_c$. The chart is analogous to Figs. 1.3-2 and 8.2-2; its use is illustrated below.

In the absence of other information, it has been suggested that Fig. 16.3-1 may be used to estimate $\mathcal{D}_{AB}$ if p_c and T_c are replaced by the pseudocritical values p_c' and T_c' given in Eqs. 1.3-3 and 4. Very few data are available to test this procedure, and therefore it has to be regarded as provisional.

² Detailed comparisons of this and other estimation methods with experimental data are given by R. C. Reid and T. K. Sherwood, *The Properties of Gases and Liquids*, McGraw-Hill, New York (1958), pp. 267-276.

Example 16.3-1. Estimation of Mass Diffusivity at Low Density

Estimate $\mathcal{D}_{AB}$ for the system argon-oxygen at 293.2° K and 1 atm total pressure.

Solution. The properties needed for Eq. 16.3-1 (see Table B-1) are

	<i>M</i>	<i>T_c</i>	<i>p_c</i>
<i>A</i> (argon)	39.94	151.2	48.0
<i>B</i> (oxygen)	32.00	154.4	49.7

Therefore

$$(p_{cA}p_{cB})^{1/2} = (48.0 \times 49.7)^{1/2} = 13.36$$

$$(T_{cA}T_{cB})^{5/12} = (151.2 \times 154.4)^{5/12} = 66.08$$

$$\left(\frac{1}{M_A} + \frac{1}{M_B}\right)^{1/2} = \left(\frac{1}{39.94} + \frac{1}{32.00}\right)^{1/2} = 0.2372$$

$$a \left(\frac{T}{\sqrt{T_{cA}T_{cB}}}\right)^b = 2.745 \times 10^{-4} \left(\frac{293.2}{\sqrt{151.2 \times 154.4}}\right)^{1.823} = 9.01 \times 10^{-4}$$

Substitution of these numerical values into Eq. 16.3-1 gives

$$(1.0)\mathcal{D}_{AB} = (9.01 \times 10^{-4})(13.36)(66.08)(0.2372)$$

$$\mathcal{D}_{AB} = 0.189 \text{ cm}^2 \text{ sec}^{-1}$$

The measured value (Table 16.2-2) is 0.20 cm² sec⁻¹. See Example 16.4-1 for a method based on kinetic theory.

Example 16.3-2. Estimation of Mass Diffusivity at High Density

Estimate $\mathcal{D}_{AB}$ for a mixture of 80 mole per cent methane and 20 mole per cent ethane at 2000 psia and 104° F (136 atm and 313° K). The experimental value of $(p\mathcal{D}_{AB})^0$ at 293° K is 0.163 atm cm² sec⁻¹.

Solution. First correct the given value of $(p\mathcal{D}_{AB})^0$ to the desired temperature, using Eq. 16.3-1 with the nonpolar constants:

$$(p\mathcal{D}_{AB})^0 \text{ at } 313^\circ \text{ K} = 0.163 \left(\frac{313}{293}\right)^{1.823} = 0.184 \text{ atm cm}^2 \text{ sec}^{-1}$$

The critical properties from Table B-1 in the Appendix are

methane:	<i>T_c</i> = 190.7° K,	<i>p_c</i> = 45.8 atm
ethane:	<i>T_c</i> = 305.4° K,	<i>p_c</i> = 48.2 atm

The pseudocritical properties according to Eqs. 1.3-4 and 5 are

$$p_c' = \sum_{i=1}^n x_i p_{ci} = (0.80 \times 45.8) + (0.20 \times 48.2) = 46.3 \text{ atm}$$

$$T_c' = \sum_{i=1}^n x_i T_{ci} = (0.80 \times 190.7) + (0.20 \times 305.4) = 213.6^\circ \text{ K}$$

The pseudo-reduced pressure and temperature are

$$\frac{p}{p_c'} = \frac{136}{46.3} = 2.94$$

$$\frac{T}{T_c'} = \frac{313}{213.6} = 1.47$$

From Fig. 16.3-1 at these reduced conditions we obtain $p\mathcal{D}_{AB}/(p\mathcal{D}_{AB})^0 \doteq 0.73$. Then

$$\begin{aligned}\mathcal{D}_{AB} &= \frac{(p\mathcal{D}_{AB})^0}{p} \frac{p\mathcal{D}_{AB}}{(p\mathcal{D}_{AB})^0} = \frac{0.184}{136} \quad (0.73) \\ &= 9.9 \times 10^{-4} \text{ cm}^2 \text{ sec}^{-1}\end{aligned}$$

An observed value of $\mathcal{D}_{AB} = 8.4 \times 10^{-4} \text{ cm}^2 \text{ sec}^{-1}$ has been reported at these conditions.³ If $(p\mathcal{D}_{AB})^0$ is predicted by the Chapman-Enskog theory given in §16.4, $\mathcal{D}_{AB}$ is predicted to be $8.8 \times 10^{-4} \text{ cm}^2 \text{ sec}^{-1}$, which is unexpectedly good agreement.

§16.4 THEORY OF ORDINARY DIFFUSION IN GASES AT LOW DENSITY

The mass diffusivity $\mathcal{D}_{AB}$ for binary mixtures of nonpolar gases is predictable within about 5 per cent by kinetic theory. Again, as in §§1.4 and 8.3, we begin with a simplified derivation to illustrate the mechanisms involved and then present the more accurate results of the Chapman-Enskog theory.

Consider a large body of gas containing two molecular species A and A^* , both species having the same mass m_A and the same size and shape. Such a pair is closely approximated by two heavy molecules containing different isotopes, such as U^{235}F_6 and U^{238}F_6 . We wish to determine the mass diffusivity $\mathcal{D}_{AA^*}$ in terms of the molecular properties on the assumption that the molecules are rigid spheres of diameter d_A .

As before, we shall use the following results of kinetic theory for a pure¹ rigid-sphere gas at low density in which the temperature, pressure, and velocity gradients are small:

$$\bar{u} = \sqrt{\frac{8KT}{\pi m}} = \text{mean molecular speed relative to } u \text{ or } v^* \quad (16.4-1)$$

$$Z = \frac{1}{2}n\bar{u} = \text{wall collision frequency per unit area in stationary gas} \quad (16.4-2)$$

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 n} = \text{mean free path} \quad (16.4-3)$$

³ V. J. Berry, Jr. and R. C. Koeller, Diffusion in Compressed Binary Gaseous Systems, Preprint 45, A.I.Ch.E. 40th National Meeting, 1959.

¹ The use of the expressions for a pure gas is permissible here because the mechanical properties of A and A^* are the same.

The molecules reaching any plane in the gas have, on the average, had their last collision at a distance a from the plane, where

$$a = \frac{2}{3}\lambda \quad (16.4-4)$$

To determine the diffusivity $\mathcal{D}_{AA^*}$, we consider the motion of species A in the y -direction under a concentration gradient dx_A/dy (see Fig. 16.4-1), when the mixture moves at a finite velocity v_y^* throughout. The temperature T and total molar concentration c are assumed constant. We assume that

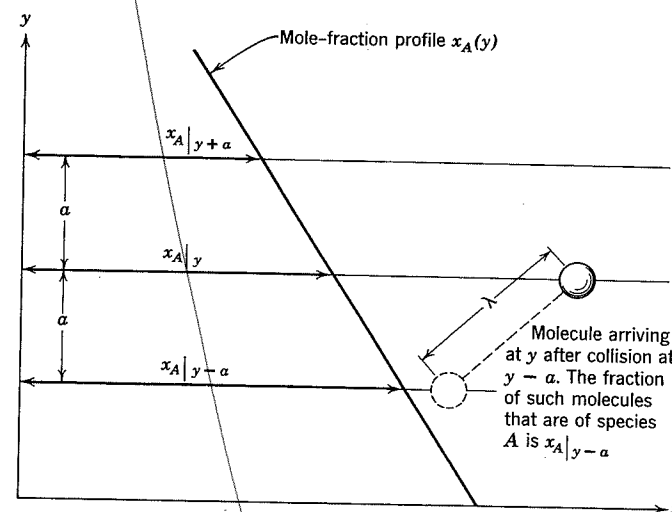


Fig. 16.4-1. Molecular transport of species A from plane at $(y - a)$ to plane at y .

Eqs. 16.4-1, 2, 3, and 4 remain valid in this nonequilibrium situation. The molar flux, N_{Ay} , of species A across any plane of constant y is found by counting the molecules of A that cross unit area of the plane in the positive y -direction and subtracting the number that cross in the negative y -direction. Thus

$$N_{Ay} = \frac{1}{N} \{ nx_A v_y^* |_y + \frac{1}{2} nx_A \bar{u} |_y - \frac{1}{2} nx_A \bar{u} |_{y+a} \} \quad (16.4-5)$$

In writing the last two terms, it is assumed that the concentration profile $x_A(y)$ is essentially linear for a distance of several mean free paths. As a further consequence of this assumption we may write

$$x_A|_{y-a} = x_A|_y - \frac{2}{3}\lambda \frac{dx_A}{dy} \quad (16.4-6)$$

$$x_A|_{y+a} = x_A|_y + \frac{2}{3}\lambda \frac{dx_A}{dy} \quad (16.4-7)$$

By combining Eqs. 16.4-5, 6, and 7 and noting that $cv^* = N_A + N_{A^*}$ (see Table 16.1-3, Eq. M), we get

$$N_{Ay} = x_A(N_{Ay} + N_{A^*y}) - \frac{1}{3} c \bar{u} \lambda \frac{dx_A}{dy} \quad (16.4-8)$$

This equation corresponds to the y -component of Fick's law (Eq. 16.2-2) with the following approximate value for $\mathcal{D}_{AB}$:

$$\mathcal{D}_{AB} = \frac{1}{3} \bar{u} \lambda \quad (16.4-9)$$

Evaluation of $\bar{u}$ and λ from Eqs. 16.4-1 and 3 and use of the ideal-gas law $p = cRT = n\kappa T$ gives

$$\mathcal{D}_{AA^*} = \frac{2}{3} \left(\frac{\kappa^3}{\pi^3 m_A} \right)^{1/2} \frac{T^{3/2}}{p d_A^2} \quad (16.4-10)$$

which represents the mass diffusivity of a mixture of two species of rigid spheres of identical mass and diameter. The calculation of $\mathcal{D}_{AB}$ for rigid spheres of unequal mass and diameter is considerably more difficult; the corresponding result² is

$$\mathcal{D}_{AB} = \frac{2}{3} \left(\frac{\kappa^3}{\pi^3} \right)^{1/2} \left(\frac{1}{2m_A} + \frac{1}{2m_B} \right)^{1/2} \frac{T^{3/2}}{p \left(\frac{d_A + d_B}{2} \right)^2} \quad (16.4-11)$$

Equations 16.4-10 and 11 predict that the mass diffusivity varies inversely with pressure; this prediction agrees well with experimental data up to about 10 atm for many gas mixtures. (See Fig. 16.3-1.) The predicted temperature dependence is too weak, just as it was for viscosity and thermal conductivity, but it is qualitatively correct.

The foregoing development is illustrative only; for accurate results, the Chapman-Enskog kinetic theory should be used. The Chapman-Enskog formulas for viscosity and thermal conductivity were given in §§1.4 and 8.3; the corresponding formula for $c\mathcal{D}_{AB}$ for the gaseous state at low density is:^{3,4}

$$c\mathcal{D}_{AB} = 2.2646 \times 10^{-5} \frac{\sqrt{T \left(\frac{1}{M_A} + \frac{1}{M_B} \right)}}{\sigma_{AB}^2 \Omega_{\mathcal{D},AB}} \quad (16.4-12)$$

² A similar result is given by R. D. Present, *Kinetic Theory of Gases*, McGraw-Hill, New York (1958), p. 55.

³ S. Chapman and T. G. Cowling, *Mathematical Theory of Non-Uniform Gases*, Second Edition, Cambridge University Press (1951), Chapters 10 and 14.

⁴ J. O. Hirschfelder, C. F. Curtiss, and R. B. Bird, *Molecular Theory of Gases and Liquids*, Wiley, New York (1954), p. 539.

If we approximate c by the ideal-gas law $c = p/RT$, this becomes

$$\mathcal{D}_{AB} = 0.0018583 \frac{\sqrt{T^3 \left(\frac{1}{M_A} + \frac{1}{M_B} \right)}}{p \sigma_{AB}^2 \Omega_{\mathcal{D},AB}} \quad (16.4-13)$$

in which $\mathcal{D}_{AB}$ [=] cm² sec⁻¹, c [=] g-moles cm⁻³, T [=] °K, p [=] atm, σ_{AB} [=] Ångström units, and $\Omega_{\mathcal{D},AB}$ is a dimensionless function of the temperature and of the intermolecular potential field for one molecule of A and one of B . It is convenient to approximate this potential field by the Lennard-Jones function:

$$\varphi_{AB}(r) = 4\epsilon_{AB} \left[\left(\frac{\sigma_{AB}}{r} \right)^{12} - \left(\frac{\sigma_{AB}}{r} \right)^6 \right] \quad (16.4-14)$$

Table B-2 gives $\Omega_{\mathcal{D},AB}$ for this potential energy model as a function of $\kappa T/\epsilon_{AB}$. From these results one can compute that $\mathcal{D}_{AB}$ increases roughly as the 2.0 power of T at low temperatures and as the 1.65 power of T at very high temperatures. (Compare Eq. 16.3-1.) For rigid spheres, $\Omega_{\mathcal{D},AB}$ would be unity at all temperatures and a result analogous to Eq. 16.4-11 would be obtained.

The Lennard-Jones parameters σ_{AB} and ϵ_{AB} could, in principle, be determined directly from accurate measurements of $\mathcal{D}_{AB}$ over a wide range of temperature; with somewhat greater difficulty, other measured properties of mixtures of A and B could be used. Suitable data are, however, extremely rare; therefore, one usually has to estimate σ_{AB} and ϵ_{AB} . It turns out that fairly satisfactory estimates can be made for nonpolar, nonreacting⁵ molecule pairs by combining the Lennard-Jones parameters of species A and B empirically:⁶

$$\sigma_{AB} = \frac{1}{2}(\sigma_A + \sigma_B) \quad (16.4-15)$$

$$\epsilon_{AB} = \sqrt{\epsilon_A \epsilon_B} \quad (16.4-16)$$

In this way it is possible to predict measured values of $\mathcal{D}_{AB}$ within an average deviation of about 6 per cent by use of viscosity data on the pure species A and B or within about 10 per cent if the Lennard-Jones parameters for A and B are estimated from boiling-point data by means of Eqs. 1.4-14 and 15.⁷

For isotopic pairs, $\sigma_{AA^*} = \sigma_A = \sigma_{A^*}$ and $\epsilon_{AA^*} = \epsilon_A = \epsilon_{A^*}$; that is, the intermolecular force fields for the various pairs A - A , A^* - A^* , and A - A^*

⁵ By nonreacting we mean here that A and B , in a bimolecular collision, do not react. Such a collision, of course, rules out the presence of any catalytic agent, so that molecule-pairs that react only catalytically are nonreacting pairs in the present discussion.

⁶ J. O. Hirschfelder, R. B. Bird, and E. L. Spotz, *Chem. Revs.*, **44**, 205-231 (1949).

⁷ R. C. Reid and T. K. Sherwood, *The Properties of Gases and Liquids*, McGraw-Hill, New York (1958), pp. 274-5.

are identical and the parameters σ_A and ϵ_A may be obtained from viscosity data on pure A . If, in addition, M_A is large, Eq. 16.4-12 simplifies to

$$c\mathcal{D}_{AA^*} = 3.2027 \times 10^{-5} \frac{\sqrt{T/M_A}}{\sigma_A^2 \Omega_{\mathcal{D},A}} \quad (16.4-17)$$

An approximate equation for $\mathcal{D}_{AA^*}$ was derived earlier. (Compare Eq. 16.4-10.)

Comparison of Eqs. 16.4-17 and 1.4-18 shows that the self-diffusivity $\mathcal{D}_{AA^*}$ and the viscosity μ are related for heavy isotopic gas-pairs at low density:

$$\frac{\mu}{\rho \mathcal{D}_{AA^*}} = \frac{\nu}{\mathcal{D}_{AA^*}} = \frac{5}{6} \frac{\Omega_{\mathcal{D},A}}{\Omega_{\mu}} \quad (16.4-18)$$

in which $\Omega_{\mathcal{D},A}$ and Ω_{μ} are nearly equal functions of $\kappa T/\epsilon_A$. (See Table B-2.) Thus the kinematic viscosity ν and the self-diffusivity $\mathcal{D}_{AA^*}$ are of the same order of magnitude for gases at low density. The relation between ν and the binary diffusivity $\mathcal{D}_{AB}$ is not so simple because $\nu = \mu/\rho$ may vary considerably with composition, but the ratio $\mu/\rho \mathcal{D}_{AB}$ lies between 0.2 and 5.0 for most gas-pairs. The quantity $\mu/\rho \mathcal{D}_{AB} = \nu/\mathcal{D}_{AB}$ is known as the *Schmidt number*; we shall encounter this quantity repeatedly in problems of diffusion in flow systems just as we encountered the Prandtl number $\hat{C}_p \mu/k = \nu/\alpha$ in problems of heat conduction in flow systems.

Equations 16.4-12, 13, 17, and 18 were derived for monatomic nonpolar gases but have been found to work well for polyatomic nonpolar gases as well. In addition, these equations may be used to predict $\mathcal{D}_{AB}$ for interdiffusion of a polar gas and a nonpolar gas⁸ by using combining laws different from Eqs. 16.4-15 and 16.

Example 16.4-1. Computation of Mass Diffusivity at Low Density

Predict the value of $\mathcal{D}_{AB}$ for mixtures of argon (A) and oxygen (B) at 293.2° K and 1 atm total pressure.

Solution. From Table B-1 we obtain the following constants:

$$M_A = 39.944; \quad \sigma_A = 3.418 \text{ Å}; \quad \frac{\epsilon_A}{K} = 124^\circ \text{ K}$$

$$M_B = 32.000; \quad \sigma_B = 3.433 \text{ Å}; \quad \frac{\epsilon_B}{K} = 113^\circ \text{ K}$$

The parameters σ_{AB} and ϵ_{AB}/K for collisions of argon with oxygen may be estimated by means of Eqs. 16.4-15 and 16:

$$\sigma_{AB} = (\frac{1}{2})(3.418 + 3.433) = 3.426 \text{ Å}$$

$$\frac{\epsilon_{AB}}{K} = \sqrt{(124)(113)} = 118.5^\circ \text{ K}$$

⁸ J. O. Hirschfelder, C. F. Curtiss, and R. B. Bird, *loc. cit.*, §8.6b.

This gives $\kappa T/\epsilon_{AB} = 293.2/118.5 = 2.47$, and from Table B-2 we then obtain $\Omega_{\mathcal{D},AB} = 1.003$. Substitution of the foregoing values into Eq. 16.4-13 gives

$$\begin{aligned} \mathcal{D}_{AB} &= 0.0018583 \frac{\sqrt{T^3 \left(\frac{1}{M_A} + \frac{1}{M_B} \right)}}{\rho \sigma_{AB}^2 \Omega_{\mathcal{D},AB}} \\ &= 0.0018583 \frac{\sqrt{(293.2)^3 \left(\frac{1}{39.944} + \frac{1}{32.00} \right)}}{(1.00)(3.426)^2 (1.003)} \\ &= 0.188 \text{ cm}^2 \text{ sec}^{-1} \end{aligned}$$

This compares favorably with the measured value of 0.20 cm² sec⁻¹ given in Table 16.2-2.

§16.5 THEORIES OF ORDINARY DIFFUSION IN LIQUIDS

In the absence of rigorous theory for diffusion in liquids, there are two rough theories that have been useful in making order of magnitude of calculations: the *hydrodynamical theory* and the *Eyring theory*. (We have already given some results of the theories of Eyring and co-workers in connection with the calculation of viscosity and thermal conductivity of liquids.)

The *hydrodynamical theory* takes as its starting point the Nernst-Einstein equation,¹ which states that the diffusivity of a single particle or solute molecule of A through a stationary medium B is

$$\mathcal{D}_{AB} = \kappa T \frac{u_A}{F_A} \quad (16.5-1)$$

in which u_A/F_A is the "mobility" of the particle of A (that is, the steady-state velocity attained by the particle under the action of a unit force). A relation between force and velocity for a rigid sphere moving in "creeping flow" (that is, $\text{Re} \ll 1$) may be obtained from hydrodynamics. If the possibility of "slip" at the sphere-fluid interface is taken into account, then²

$$F_A = 6\pi\mu_B u_A R_A \left(\frac{2\mu_B + R_A \beta_{AB}}{3\mu_B + R_A \beta_{AB}} \right) \quad (16.5-2)$$

in which μ_B is the viscosity of the pure solvent, R_A is the radius of the diffusing particle, and β_{AB} is the *coefficient of sliding friction*. It has been pointed out³ that there are two limiting cases of interest:

¹ F. Daniels and R. A. Alberty, *Physical Chemistry*, Wiley, New York (1955), p. 650. See also Eq. 18.4-14a.

² H. Lamb, *Hydrodynamics*, Dover, New York, §337.

³ R. Fürth, in *Handbuch der physikalischen und technischen Mechanik*, Barth, Leipzig (1931), Vol. 7, p. 635.

a. If there is no tendency for the fluid to slip at the surface of the diffusing particle ($\beta_{AB} = \infty$), then Eq. 16.5-2 becomes Stokes's law. (See §2.6.)

$$F_A = 6\pi\mu_B u_A R_A \quad (16.5-3)$$

and substitution into Eq. 16.5-1 gives

$$\frac{\mathcal{D}_{AB}\mu_B}{KT} = \frac{1}{6\pi R_A} \quad (16.5-4)$$

which is usually called the *Stokes-Einstein equation*. Equation 16.5-4 has been shown to be fairly good for describing the diffusion of *large spherical particles* or *large spherical molecules*, under which conditions the solvent appears to the diffusing species as a continuum.⁴

b. If there is no tendency for the fluid to stick at the surface of the diffusing particle ($\beta_{AB} = 0$), Eq. 16.5-2 becomes

$$F_A = 4\pi\mu_B u_A R_A \quad (16.5-5)$$

and Eq. 16.5-1 becomes

$$\frac{\mathcal{D}_{AB}\mu_B}{KT} = \frac{1}{4\pi R_A} \quad (16.5-6)$$

If the molecules are all alike (that is, *self-diffusion*) and if they can be assumed to be arranged in a cubic lattice with all molecules just touching, then $2R_A$ may be set equal to $(\bar{V}_A/\tilde{N})^{1/3}$ and

$$\frac{\mathcal{D}_{AA}\mu_A}{KT} = \frac{1}{2\pi} \left(\frac{\tilde{N}}{\bar{V}_A} \right)^{1/3} \quad (16.5-7)$$

It has been shown⁵ that Eq. 16.5-7 predicts the self-diffusion data for a number of liquids within about ± 12 per cent; that comparison includes polar substances, associated substances, liquid metals, and molten sulfur.

It is thus seen that the simple hydrodynamical approach gives expressions for the diffusion coefficient for spherical molecules in dilute solution and also for the coefficient of self-diffusion. The theory further predicts that there should be a variation of $\mathcal{D}_{AB}$ with the size of the diffusing species. The hydrodynamic theory suggests that the shape of the diffusing species may well be important, since the friction factor increases by a factor of about 2 as the length-to-width ratio of a body goes from 1 to 10.

The *Eyring rate theory*⁶ attempts to explain the transport phenomena on the basis of a simple model for the liquid state. It is assumed in this theory that there is some unimolecular rate process in terms of which the diffusion

⁴ W. Sutherland, *Phil. Mag.*, **9**, 781-785 (1905).

⁵ J. C. M. Li and P. Chang, *J. Chem. Phys.*, **23**, 518-520 (1955).

⁶ S. Glasstone, K. J. Laidler, and H. Eyring, *Theory of Rate Processes*, McGraw-Hill, New York (1941), Chapter IX.

process can be described, and it is further assumed that in this process there is some configuration that can be identified as the *activated state*. The Eyring theory of reaction rates is applied to this elementary process.

On the assumption of a cubic lattice configuration, this theory gives the following relation between the self-diffusion coefficient and the coefficient of viscosity:

$$\frac{\mathcal{D}_{AA}\mu_A}{KT} = \left(\frac{\tilde{N}}{\bar{V}_A} \right)^{1/3} \quad (16.5-8)$$

This relation differs by a factor of 2π from Eq. 16.5-7, the expression for the self-diffusion coefficient according to the hydrodynamic theory; but Eq. 16.5-7 has been shown to fit the experimental data better than Eq. 16.5-8. Recently Li and Chang⁷ have suggested a modification of the Eyring theory in order to account for the discrepancy.

Because of the approximate nature of the foregoing theories, a number of *empirical relations* have been proposed,⁸ one of which we mention briefly.

Wilke⁹ has developed a correlation for diffusion coefficients on the basis of the Stokes-Einstein equation. His results may be summarized by the following approximate analytical relation,¹⁰ which gives the diffusion coefficient in $\text{cm}^2 \text{sec}^{-1}$ for small concentrations of *A* in *B*:

$$\mathcal{D}_{AB} = 7.4 \times 10^{-8} \frac{(\psi_B M_B)^{1/2} T}{\mu \bar{V}_A^{0.6}} \quad (16.5-9)$$

Here $\bar{V}_A$ is the molar volume of the solute *A* in $\text{cm}^3 \text{g-mole}^{-1}$ as liquid at its normal boiling point, μ is the viscosity of the solution in centipoises, ψ_B is an "association parameter" for the solvent *B*, and *T* is the absolute temperature in $^\circ \text{K}$. Recommended values of ψ_B are 2.6 for water, 1.9 for methanol, 1.5 for ethanol, and 1.0 for benzene, ether, heptane, and other unassociated solvents. This equation is good only for dilute solutions of nondissociating solutes; for such solutions it is usually good within ± 10 per cent.

If the reader has by now concluded that little is known about the prediction of dense gas and liquid diffusivities, he is correct. There is an urgent need for experimental measurements, both for their own value and for development of future theories.

Example 16.5-1. Estimation of Mass Diffusivity for a Binary Liquid Mixture

Estimate $\mathcal{D}_{AB}$ for a dilute solution of TNT (2,4,6-trinitrotoluene) in benzene at 15°C .

⁷ Li and Chang, *loc. cit.*

⁸ See, for example, R. C. Reid and T. K. Sherwood, *The Properties of Gases and Liquids*, McGraw-Hill, New York (1958), pp. 284-298.

⁹ C. R. Wilke, *Chem. Eng. Prog.*, **45**, 218-224 (1949).

¹⁰ C. R. Wilke and P. Chang, *A.I.Ch.E. Journal*, **1**, 264-270, 1955.

Solution. Use the equation of Wilke and Chang, taking TNT as component *A* and benzene as component *B*. The required data are

$$\begin{aligned}\mu &= 0.705 \text{ cp (for solution considered as pure benzene)} \\ \bar{V}_A &= 140 \text{ cc g-mole}^{-1} \text{ for TNT}^{11} \\ \psi_B &= 1.0 \text{ for benzene} \\ M_B &= 78.11 \text{ for benzene}\end{aligned}$$

Substitution in Eq. 16.5-9 gives

$$\begin{aligned}\mathcal{D}_{AB} &= 7.4 \times 10^{-8} \frac{(1.0 \times 78.11)^{1/2}(273 + 15)}{0.705(140)^{0.6}} \\ &= 1.38 \times 10^{-5} \text{ cm}^2 \text{ sec}^{-1}\end{aligned}$$

This result may be compared with a measured value of $1.39 \times 10^{-5} \text{ cm}^2 \text{ sec}^{-1}$.

QUESTIONS FOR DISCUSSION

1. What is diffusion? What factors may cause diffusion to occur?
2. When is mass fraction equal to mole fraction?
3. Compare the mass average velocity v in Eq. 16.1-1 with the average velocity defined in §2.2.
4. Compare the significance of the fluxes N_i and J_i^* . Compare the significance of the fluxes n_i and f_i .
5. Why is it necessary to specify a reference velocity in reporting data on rates of diffusion?
6. Compare Fick's law of diffusion with Newton's law of viscosity and Fourier's law of thermal conductivity. To what extent are these three relations analogous?
7. Give the special forms of Fick's law for the flux N_A that apply to (a) equimolar counterdiffusion ($N_A = -N_B$), (b) diffusion of *A* through stagnant *B* ($N_B = 0$).
8. What are the dimensions of $\mathcal{D}_{AB}$, v , and α ?
9. Compare the order of magnitude of $\mathcal{D}_{AB}$ and of $c\mathcal{D}_{AB}$ for gases and liquids at 1 atm.
10. What is the difference in behavior of the diffusivity-concentration curve for ethanol-water and that for *n*-butanol-water in Table 16.2-3?
11. By means of a small calculation, verify the statement concerning the temperature dependence of $\mathcal{D}_{AB}$ just after Eq. 16.4-14.
12. What empirical method is available for predicting the intermolecular force field between a pair of dissimilar nonpolar molecules?
13. What bulk physical properties are related by the Stokes-Einstein equation? Verify that the equation is dimensionally consistent.
14. Compare the relationship between $\mathcal{D}_{AB}$ and μ for gases with that for liquids.
15. What percentage of error in the predicted value of $\mathcal{D}_{AB}$ from Eq. 16.4-13 would result from a 5 per cent error in σ_{AB} ? In T ? In ϵ_{AB}/K ?
16. In binary diffusion, does the statement that the mole fraction gradient ∇x_A is not zero imply that the molar flux N_A is not zero? Explain.

¹¹ Estimated by the atomic-volume method of Schroeder as quoted by R. C. Reid and T. K. Sherwood, *The Properties of Gases and Liquids*, McGraw-Hill, New York (1958), p. §1.

PROBLEMS

16.A₁ Prediction of Mass Diffusivity for a Gas-Pair at Low Density

Predict $\mathcal{D}_{AB}$ for the methane-ethane system at 104° F and 1 atm, using the Chapman-Enskog theory. Compare your result with the experimental value given in Example 16.3-2.
Answer: $0.166 \text{ cm}^2 \text{ sec}^{-1}$

16.B₁ Prediction of Mass Diffusivity at Low Density from Critical Properties

Predict $\mathcal{D}_{AB}$ for the situation of Problem 16.A by the following two methods:

- a. The Slattery equation, Eq. 16.3-1.
 - b. The Chapman-Enskog theoretical equation, Eq. 16.4-13, using critical pressures and temperatures to estimate the Lennard-Jones parameters. (See Eqs. 1.4-11 and 13.)
- Answers: a. $0.172 \text{ cm}^2 \text{ sec}^{-1}$
b. $0.156 \text{ cm}^2 \text{ sec}^{-1}$

16.C₁ Correction of Mass Diffusivity for Temperature at Low Density

A value of $\mathcal{D}_{AB} = 0.151 \text{ cm}^2 \text{ sec}^{-1}$ has been reported¹ for the system CO_2 -air at 293° K and 1 atm. Extrapolate $\mathcal{D}_{AB}$ to 1500° K by the following methods:

- a. Eq. 16.3-1.
 - b. Eq. 16.4-11.
 - c. Eq. 16.4-13.
- Answers in $\text{cm}^2 \text{ sec}^{-1}$: a. 2.96
b. 1.75
c. 2.49
Observed:¹ $2.45 \text{ cm}^2 \text{ sec}^{-1}$

16.D₁ Prediction of Mass Diffusivity for a Gas Mixture at High Density

Predict $\mathcal{D}_{AB}$ for an equimolar mixture of CO_2 and N_2 at 288.2° K and 40 atm:

- a. Use the value of $\mathcal{D}_{AB}$ at 1 atm from Table 16.2-2.
 - b. Use the Chapman-Enskog theory to predict $(p\mathcal{D}_{AB})^0$.
- Answers: a. $0.0033 \text{ cm}^2 \text{ sec}^{-1}$
b. $0.0030 \text{ cm}^2 \text{ sec}^{-1}$

16.E₁ Estimation of Mass Diffusivity for a Binary Liquid Mixture

Estimate $\mathcal{D}_{AB}$ for acetic acid in dilute aqueous solution at 12.5° C by using Eq. 16.5-9. The density of acetic acid at its normal boiling point is 0.937 g cm^{-3} . (An experimental value of $0.91 \pm 0.04 \times 10^{-5} \text{ cm}^2 \text{ sec}^{-1}$ has been reported.²)

16.F₁ Correction of Mass Diffusivity for Temperature for a Binary Liquid Mixture

The value of $\mathcal{D}_{AB}$ for a dilute solution of methanol in water at 15° C is $1.28 \times 10^{-5} \text{ cm}^2 \text{ sec}^{-1}$. Estimate $\mathcal{D}_{AB}$ for the same solution at 100° C, using the Wilke-Chang equation.
Answer: $6.7 \times 10^{-5} \text{ cm}^2 \text{ sec}^{-1}$

16.G₁ Correction of Mass Diffusivity for Temperature for a Dense Gas Mixture

The observed value of $\mathcal{D}_{AB}$ for a mixture of 80 mole per cent methane and 20 mole per cent ethane at 104° F and 2000 psia is $8.4 \times 10^{-4} \text{ cm}^2 \text{ sec}^{-1}$. Predict $\mathcal{D}_{AB}$ for the same mixture at 2000 psia and 171° F:

- a. Use Eq. 16.3-1 and Fig. 16.3-1.
- b. Use Eq. 16.4-13 and Fig. 16.3-1.

Answer: a. $1.19 \times 10^{-3} \text{ cm}^2 \text{ sec}^{-1}$

¹ Ts. M. Klibanova, V. V. Pomerantsev, and D. A. Frank-Kamenetskii, *J. Tech. Phys. (U.S.S.R.)*, **12**, 14-30 (1942), as quoted by C. R. Wilke and C. Y. Lee, *Ind. Eng. Chem.*, **47**, 1253 (1955).

² *International Critical Tables*, McGraw-Hill, New York (1929) Vol. V, p. 70.

16.H₂ Relations Among the Fluxes and Concentrations in a Binary System

Prove Eqs. K , L , and Z in Table 16.1-3, and O in Table 16.1-1, using only the definitions of the concentrations, velocities, and fluxes.

16.I₂ Equivalence of Various Forms of Fick's Law

a. Prove that all of the forms of Fick's first law given in Table 16.2-1 are equivalent, that is, that the value of $\mathcal{D}_{AB}$ is identical in all seven equations. Any relations given in §16.1 may be used.

b. Use Eq. 16.2-1 and the relations tabulated in §16.1 to prove that $\mathcal{D}_{AB} = \mathcal{D}_{BA}$.

16.J₃ Determination of Collision Parameters from Diffusivity Data

a. Use the following data^a for the system CO₂-air at 1 atm pressure to determine σ_{AB} and ϵ_{AB}/k directly, that is, without using the values of σ and ϵ/k for the separate species:

$T, ^\circ K$	293	400	600	800	1000
$\mathcal{D}_{AB}, \text{cm}^2 \text{sec}^{-1}$	0.151	0.273	0.555	0.915	1.32

It is suggested that the data be plotted as $\log(T^{3/2}/\mathcal{D}_{AB})$ versus $\log T$ on a thin sheet of graph paper and that a plot of $\Omega_{\mathcal{D},AB}$ vs. kT/ϵ_{AB} be made on a separate sheet to the same fineness of scale. Then one can superimpose the data plot on the $\Omega_{\mathcal{D}}$ -function plot and from the scales of the two overlapping plots determine the numerical ratios $T/(kT/\epsilon_{AB})$ and $(T^{3/2}/\mathcal{D}_{AB})/\Omega_{\mathcal{D},AB}$. These two ratios, together with Eq. 16.4-12, may be used to solve for the two parameters σ_{AB} and ϵ_{AB}/k .

b. Estimate σ_{AB} and ϵ_{AB}/k by the procedure of §16.4.

16.K₃ Mass Flux with Respect to Volume-Average Velocity

Let the *volume-average velocity* be defined by

$$v^{\blacksquare} = \sum_i \rho_i v_i (\bar{V}_i / M_i) \quad (16.K-1)$$

in which $\bar{V}_i$ is the partial molal volume of component i and M_i is its molecular weight. Let the mass flux with respect to the volume average velocity then be

$$j_i^{\blacksquare} = \rho_i (v_i - v^{\blacksquare}) \quad (16.K-2)$$

Show that for a binary system composed of species A and B

$$j_A^{\blacksquare} = j_A \rho (\bar{V}_B / M_B) \quad (16.K-3)$$

and that Fick's first law assumes the form

$$j_A^{\blacksquare} = -\mathcal{D}_{AB} \nabla \rho_A \quad \text{for constant } T \text{ and } p. \quad (16.K-4)$$

^a From Footnote 1, Problem 16.C.

Concentration Distributions in Solids and in Laminar Flow

In Chapter 2 it was shown how a number of simple viscous flow problems can be set up and solved by making a shell *momentum balance*. In Chapter 9 it was further shown how heat-conduction problems can be handled by means of a shell *energy balance*. In this chapter we show how elementary diffusion problems may be formulated by a shell *mass balance*. The procedure used here is essentially the same as that used previously:

a. A mass balance is made over a thin shell perpendicular to the direction of mass transport, and this shell balance leads to a first-order differential equation, which may be solved to get the mass-flux distribution.

b. Into this expression we insert the relation between mass flux and concentration gradient, and this results in a second-order differential equation for the concentration profile. The integration constants that appear in the resulting expression are determined by the boundary conditions that specify the concentration or the mass flux at the bounding surfaces.

In Chapter 16 it was pointed out that several kinds of mass fluxes are in common use. For the sake of simplicity, we shall use the flux N_A exclusively, that is, the number of moles of A that go through a unit area in unit time,

the unit area being fixed in space. We shall relate the molar flux to the concentration gradient (see Eq. 16.2-2) by

$$N_{Az} = \underbrace{-c \mathcal{D}_{AB} \frac{\partial x_A}{\partial z}}_{\text{flux with respect to fixed axes}} + \underbrace{x_A(N_{Az} + N_{Bz})}_{\text{flux resulting from diffusion}} \quad (17.0-1)$$

{ flux resulting from total molar bulk flow }

for two-component systems. Keep in mind that N_{Az} is the z -component of the vector N_A . Before Eq. 17.0-1 can be used, we shall have to eliminate N_{Bz} . This can be done only if something is known beforehand about the ratio N_{Bz}/N_{Az} . In each of the problems discussed in this chapter we begin by specifying this ratio on physical grounds.

In this chapter we study diffusion in both *nonreacting* and *reacting* systems. When chemical reactions occur, we distinguish between two types: *homogeneous*, in which the chemical change occurs in the entire volume of the fluid, and *heterogeneous*, in which the chemical change takes place only in a restricted region in the system, such as at the surface of a catalyst. Not only is the physical picture different for homogeneous and heterogeneous reactions, but there is also a difference in the way in which the two reactions are described. The rate of production by a homogeneous reaction appears in a source term in the differential equation obtained from the shell mass balance, just as the thermal source term appears in the shell energy balance. The rate of production by a heterogeneous reaction, on the other hand, appears not in the differential equation but rather in the boundary condition at the surface on which the reaction occurs.

In order to set up problems involving chemical reactions, some information has to be available about the rate at which the various chemical species appear or disappear by reaction. This brings us then to the subject of *chemical kinetics*, that branch of physical chemistry that deals with the mechanisms of chemical reactions and the rates at which they occur.¹ In this chapter we assume that the mechanisms of the reactions are known and that the reaction rates are describable by means of simple functions of the concentrations of the reacting species. Mention needs to be made at this point of the notation to be used for the chemical rate constants. For homogeneous reactions, the volume rate of production of species A may be given by an expression of the form

$$R_A = k_n'' c_A^n \quad (17.0-2)$$

in which R_A [=] moles $\text{cm}^{-3} \text{sec}^{-1}$ and c_A [=] moles cm^{-3} . The index n indicates the "order" of the reaction;² for a first-order reaction, k_1''' [=] sec^{-1} .

¹ For an introduction to chemical kinetics, see F. Daniels and R. A. Alberty, *Physical Chemistry*, Wiley, New York (1955), Ch. 13. A more mathematical discussion is given by E. A. Moelwyn-Hughes, *Physical Chemistry*, Pergamon Press, New York (1957).

² Not all rate expressions are of the simple form of Eq. 17.0-2. The reaction rate may depend in a more complex way on the concentrations of all species present. Similar remarks hold for Eq. 17.0-3.

For heterogeneous reactions, the rate of reaction at the catalytic surface may be specified by a relation of the form

$$N_{Az}|_{\text{surface}} = k_n'' c_A^n|_{\text{surface}} \quad (17.0-3)$$

in which N_{Az} [=] moles $\text{cm}^{-2} \text{sec}^{-1}$ and c_A [=] moles cm^{-3} . Here k_1'' [=] $\text{cm} \text{sec}^{-1}$. Note that $''$ indicates a rate constant related to a volume source and $'$ indicates a rate constant related to a surface source.

We begin in §17.1 with a statement of the shell mass balance and the kinds of boundary conditions that may arise in solving diffusion problems. In §17.2 a discussion of diffusion in a stagnant film is given, this topic being necessary to the understanding of the film theories of the diffusional operations in chemical engineering. Then, in §§17.3 and 17.4, we give some elementary examples of diffusion with chemical reaction—both homogeneous and heterogeneous; it is hoped that these examples will illustrate the role that diffusion plays in chemical kinetics and the important fact that there is generally a difference between the rate of the chemical reaction and the rate of the combined diffusion-reaction process. In §17.5 we turn our attention to the subject of forced-convection mass transfer—that is, diffusion superimposed on a flow field; we could also have included for the sake of completeness a section on free-convection mass transfer, which would have paralleled the discussion on free-convection heat transfer in Chapter 9. Finally, in the last section, §17.6, we discuss diffusion in porous catalysts.

§17.1 SHELL MASS BALANCES: BOUNDARY CONDITIONS

The diffusion problems in this chapter are solved by making mass balances for a specific chemical species over a thin shell of solid or fluid. Having selected an appropriate system, the law of conservation of mass is written in the form

$$\left\{ \begin{array}{c} \text{rate of} \\ \text{mass of} \\ A \text{ in} \end{array} \right\} - \left\{ \begin{array}{c} \text{rate of} \\ \text{mass of} \\ A \text{ out} \end{array} \right\} + \left\{ \begin{array}{c} \text{rate of production} \\ \text{of mass of } A \text{ by} \\ \text{homogeneous chemical} \\ \text{reaction} \end{array} \right\} = 0 \quad (17.1-1)$$

The conservation statement may, of course, also be expressed in terms of moles. The chemical species A may enter or leave the system by means of diffusion and by virtue of the over-all motion of the fluid. Also species A may be produced or destroyed by homogeneous chemical reactions.

After a balance is made on a shell of finite thickness, according to Eq. 17.1-1, we then let the dimensions of the system become infinitesimally small. As a result of this process, a differential equation is generated, the solution of which gives the distribution of species A in the system.

When the differential equation has been integrated, constants of integration appear, which have to be evaluated by use of boundary conditions.

The boundary conditions used are very similar to those used in energy transfer. (See §9.1.)

- The concentration at a surface can be specified; for example, $x_A = x_{A0}$.
- The mass flux at a surface can be specified (if the ratio N_A/N_B is known, this is tantamount to giving the concentration gradient); for example, $N_A = N_{A0}$.
- If diffusion is occurring in a solid, it may happen that at the solid surface substance A is lost to a surrounding fluid stream according to the relation

$$N_{A0} = k_c(c_{A0} - c_{Af}) \quad (17.1-2)$$

in which N_{A0} is the mass flux at the surface, c_{A0} is the surface concentration, c_{Af} is the concentration in the fluid stream, and the proportionality constant k_c is a "mass-transfer coefficient"; the methods of correlating mass-transfer coefficients are discussed in Chapter 21. Equation 17.1-2 is analogous to "Newton's law of cooling" defined in Eq. 9.1-2.

- The rate of chemical reaction at the surface can be specified. For example, if substance A disappears at a surface by a first-order chemical reaction, $N_{A0} = k_1 c_A$; that is, the rate of disappearance at a surface is proportional to the surface concentration, the proportionality constant k_1 being a first-order chemical rate constant.

In the following sections we shall see how these various boundary conditions are used.

§17.2 DIFFUSION THROUGH A STAGNANT GAS FILM

Consider the diffusion system shown in Fig. 17.2-1. Liquid A is evaporating into gas B , and we imagine that there is some device which maintains the liquid level at $z = z_1$. Right at the liquid-gas interface the gas-phase concentration of A , expressed as mole fraction, is x_{A1} . This is taken to be the gas-phase concentration of A corresponding to equilibrium¹ with the liquid at the interface; that is, x_{A1} is the vapor pressure of A divided by the total pressure, $p_A^{(vap)}/p$, provided that A and B form an ideal gas mixture. We further assume that the solubility of B in liquid A is negligible.

At the top of the tube (at $z = z_2$) a stream of gas mixture A - B of concentration x_{A2} flows past slowly; thereby the mole fraction of A at the top of the column is maintained at x_{A2} . The entire system is presumed to be held

¹ The assumption of equilibrium at the interface has been subjected to experimental test by L. N. Tung and H. G. Drickamer, *J. Chem. Phys.*, **20**, 6-12 (1952), and by R. E. Emmert and R. L. Pigford, *Chem. Eng. Prog.*, **50**, 87-93 (1954). A kinetic theory approach was made by R. W. Schrage, *A Theoretical Study of Interphase Mass Transfer*, Columbia University Press, New York (1953). These studies seem to indicate that only at very high mass-transfer rates is there a significant departure from equilibrium at the interface.

at constant temperature and pressure. Gases A and B are assumed to be ideal.

When this evaporating system attains a steady state, there is a net motion of A away from the evaporating surface and the vapor B is stationary. Hence

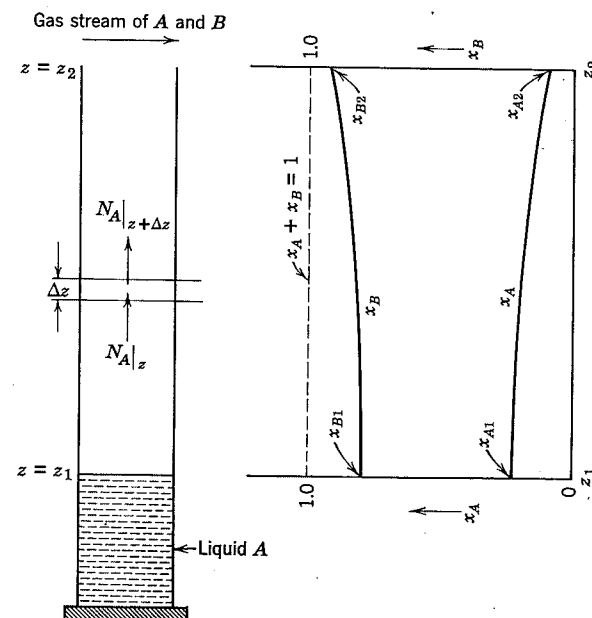


Fig. 17.2-1. Diffusion of A through B in steady state with B not in motion. Graph shows how concentration profile for B is distorted because of diffusion of A .

we can use the expression for N_{Az} given in Eq. 17.0-1 with $N_{Bz} = 0$. Solving for N_{Az} , we get

$$N_{Az} = - \frac{c \mathcal{D}_{AB}}{1 - x_A} \frac{dx_A}{dz} \quad (17.2-1)$$

A mass balance over an incremental column height Δz (see Fig. 17.2-1) states that at steady state

$$S N_{Az}|_z - S N_{Az}|_{z+\Delta z} = 0 \quad (17.2-2)$$

in which S is the cross-sectional area of the column. Division by $S \Delta z$ and taking the limit as Δz approaches zero gives

$$- \frac{dN_{Az}}{dz} = 0 \quad (17.2-3)$$

Substitution of Eq. 17.2-1 into Eq. 17.2-3 gives

$$\frac{d}{dz} \left(\frac{c\mathcal{D}_{AB}}{1-x_A} \frac{dx_A}{dz} \right) = 0 \quad (17.2-4)$$

For ideal-gas mixtures at constant temperature and pressure, c is a constant, and $\mathcal{D}_{AB}$ is very nearly independent of concentration. Hence $c\mathcal{D}_{AB}$ can be taken outside the derivative to get

$$\frac{d}{dz} \left(\frac{1}{1-x_A} \frac{dx_A}{dz} \right) = 0 \quad (17.2-5)$$

This is a second-order equation for the concentration profile expressed as mole fraction of A . Integration with respect to z gives

$$\frac{1}{1-x_A} \frac{dx_A}{dz} = C_1 \quad (17.2-6)$$

A second integration then gives

$$-\ln(1-x_A) = C_1 z + C_2 \quad (17.2-7)$$

The two constants of integration may be determined by the use of the boundary conditions

$$\text{B.C. 1:} \quad \text{at } z = z_1, \quad x_A = x_{A1} \quad (17.2-8)$$

$$\text{B.C. 2:} \quad \text{at } z = z_2, \quad x_A = x_{A2} \quad (17.2-9)$$

When the constants so obtained² are substituted into Eq. 17.2-7, the following expressions for the concentration profiles are obtained:

$$\boxed{\left(\frac{1-x_A}{1-x_{A1}} \right) = \left(\frac{1-x_{A2}}{1-x_{A1}} \right)^{\frac{z-z_1}{z_2-z_1}}} \quad (17.2-10)$$

or

$$\boxed{\left(\frac{x_B}{x_{B1}} \right) = \left(\frac{x_{B2}}{x_{B1}} \right)^{\frac{z-z_1}{z_2-z_1}}} \quad (17.2-11)$$

² The constants are

$$C_1 = \frac{\ln x_{B1} - \ln x_{B2}}{z_2 - z_1} = \frac{\ln(x_{B1}/x_{B2})}{z_2 - z_1}; \quad C_2 = \frac{z_1 \ln x_{B2} - z_2 \ln x_{B1}}{z_2 - z_1}$$

To get Eq. 17.2-10, first rewrite C_2 thus:

$$C_2 = \frac{z_1 \ln x_{B2} - (z_2 - z_1) \ln x_{B1} - z_1 \ln x_{B1}}{z_2 - z_1} = -\frac{z_1}{z_2 - z_1} \ln \frac{x_{B1}}{x_{B2}} - \ln x_{B1}$$

These concentration distributions are depicted in Fig. 17.2-1. An examination of these curves shows that the slope dx_A/dz is not constant with respect to z , although the molar flux N_{Az} is.

Although the concentration profiles are helpful in picturing the diffusion process, in engineering calculations it is usually the average concentration or the mass flux at some surface that is of interest. For example, the average concentration of B in the region between $z = z_1$ and $z = z_2$ is

$$\frac{x_{B,\text{avg}}}{x_{B1}} = \frac{\int_{z_1}^{z_2} (x_B/x_{B1}) dz}{\int_{z_1}^{z_2} dz} = \frac{\int_0^1 (x_{B2}/x_{B1})^\zeta d\zeta}{\int_0^1 d\zeta} = \frac{(x_{B2}/x_{B1})^\zeta}{\ln(x_{B2}/x_{B1})} \bigg|_0^1 \quad (17.2-12)$$

whence

$$x_{B,\text{avg}} = \frac{x_{B2} - x_{B1}}{\ln(x_{B2}/x_{B1})} \quad (17.2-13)$$

That is, the average value of x_B is the logarithmic mean of the terminal values, $(x_B)_{\ln}$. In the foregoing ζ is a reduced length $(z - z_1)/(z_2 - z_1)$.

The rate of mass transfer at the liquid-gas interface—that is, the rate of evaporation—is obtained by using Eq. 17.2-1:

$$N_{Az}|_{z=z_1} = -\frac{c\mathcal{D}_{AB}}{1-x_{A1}} \frac{dx_A}{dz} \bigg|_{z=z_1} = +\frac{c\mathcal{D}_{AB}}{x_{B1}} \frac{dx_B}{dz} \bigg|_{z=z_1} = \frac{c\mathcal{D}_{AB}}{(z_2 - z_1)} \ln \left(\frac{x_{B2}}{x_{B1}} \right) \quad (17.2-14)$$

Equations 17.2-13 and 17.2-14 may be combined to give an alternative expression for the mass-transfer rate:

$$N_{Az}|_{z=z_1} = \frac{c\mathcal{D}_{AB}}{(z_2 - z_1)(x_B)_{\ln}} (x_{A1} - x_{A2}) \quad (17.2-15)$$

This expression shows how the rate of mass transfer is related to a characteristic concentration driving force $x_{A1} - x_{A2}$. Equations 17.2-14 and 15 may also be expressed in terms of total pressure and partial pressures:

$$N_{Az}|_{z=z_1} = \frac{(p\mathcal{D}_{AB}/RT)}{(z_2 - z_1)} \ln \frac{p_{B2}}{p_{B1}} = \frac{(p\mathcal{D}_{AB}/RT)}{(z_2 - z_1)(p_B)_{\ln}} (p_{A1} - p_{A2}) \quad (17.2-15a)$$

in which $(p_B)_{\ln}$ is the logarithmic mean of p_{B1} and p_{B2} defined analogously to $(x_B)_{\ln}$ in Eq. 17.2-15.

The results of this section have been used for experimental determination of gas diffusivities.³ Furthermore, these results find use in the "film theories"

³ See, for example, C. Y. Lee and C. R. Wilke, *Ind. Eng. Chem.*, **46**, 2381-2387 (1954); T. K. Sherwood and R. L. Pigford, *Absorption and Extraction*, McGraw-Hill (1952).

of mass transfer. (See §21.5.) In Fig. 17.2-2 a solid or liquid surface is shown along which a gas is flowing. Near the surface is a slowly moving film through which A diffuses; z_1 is taken to be the solid-gas or liquid-gas interface, and z_2 is taken to be the outer limit of the gas "film" through which the diffusion occurs. In this "model" it is assumed that there is a sharp transition between a stagnant film and a well-mixed fluid in which concentration

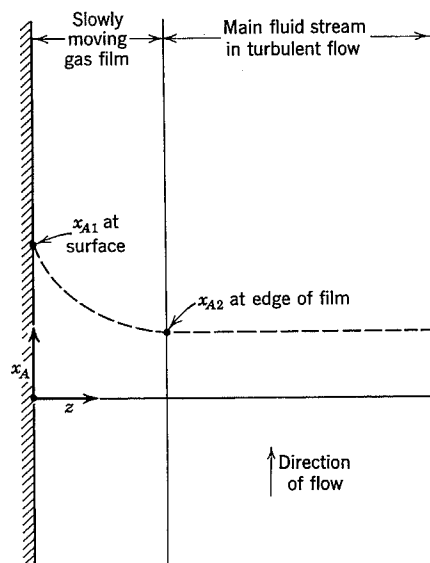


Fig. 17.2-2. Film model for mass transfer; component A is diffusing from the surface into the gas stream.

gradients are negligible. Although this model is physically unrealistic, it has nevertheless proven very useful as a basis for correlating mass-transfer coefficients in terms of a simple physical picture.

Example 17.2-1. Determination of Diffusivity

The diffusivity of the gas-pair oxygen-carbon tetrachloride is determined by observing the steady-state evaporation of CCl_4 into a tube containing O_2 , as shown in Fig. 17.2-1. The distance between the CCl_4 liquid level and the top of the tube is $z_2 - z_1 = 17.1$ cm. The total pressure on the system is 755 mm Hg, and the temperature is 0°C . The vapor pressure of CCl_4 at that temperature is 33.0 mm Hg. The cross-sectional area of the diffusion tube is 0.82 cm^2 . If it is found that 0.0208 cm^3 of CCl_4 evaporate in a 10-hour period after steady state has been attained, what is the diffusivity of the gas-pair $\text{CCl}_4\text{--O}_2$?

Solution. First we get the molar flux N_{CCl_4} from the evaporation rate:

$$N_{\text{CCl}_4} = (0.0208\text{ cm}^3) \left(1.59 \frac{\text{g}}{\text{cm}^3} \right) \left(\frac{1}{154} \frac{\text{g-mole}}{\text{g}} \right) \left(\frac{1}{0.82\text{ cm}^2} \right) \left(\frac{1}{3.6 \times 10^4\text{ sec}} \right) \\ = 7.26 \times 10^{-9} \frac{\text{g-mole}}{\text{cm}^2\text{ sec}}$$

Then from Eq. 17.2-15a we get

$$\mathcal{D}_{\text{CCl}_4\text{--O}_2} = \frac{N_{\text{CCl}_4}(z_2 - z_1)RT}{p \ln(p_{\text{O}_2,2}/p_{\text{O}_2,1})} \\ = \frac{(7.26 \times 10^{-9})(17.1)(82.06)(273)}{\left(\frac{755}{760}\right)(2.303 \log_{10} \frac{755}{722})} \\ = 0.0636\text{ cm}^2\text{ sec}^{-1}$$

This method of determining gas-phase diffusivities suffers from several defects. The most serious is probably the cooling effect associated with the evaporation. Furthermore, there may be free-convection effects if the vapor from the evaporating liquid is lighter than the gas above the liquid. Also, because of the meniscus, the length of the diffusion path will not be uniform across the tube.

Example 17.2-2. Diffusion Through a Nonisothermal Spherical Film

a. Derive expressions for diffusion through a spherical shell that are analogous to Eq. 17.2-10 (concentration profile) and Eq. 17.2-14 (molar flux); see Fig. 17.2-3 for specification of system.

b. Extend these results to describe the diffusion in a nonisothermal film in which the temperature changes with the distance according to

$$\left(\frac{T}{T_1} \right) = \left(\frac{r}{r_1} \right)^n \quad (17.2-16)$$

in which T_1 is the temperature at $r = r_1$. Assume as a rough approximation that $\mathcal{D}_{AB}$ varies as the $\frac{3}{2}$ -power of the temperature:

$$\left(\frac{\mathcal{D}_{AB}}{\mathcal{D}_{AB,1}} \right) = \left(\frac{T}{T_1} \right)^{3/2} \quad (17.2-17)$$

in which $\mathcal{D}_{AB,1}$ is the diffusivity at $T = T_1$. Problems of this kind arise in connection with drying of droplets and diffusion through gas films near spherical catalyst pellets. (The temperature distribution in Eq. 17.2-16 has been chosen solely for mathematical simplicity. This example has been included to emphasize that in nonisothermal systems Eq. 17.0-1 is the correct starting point rather than $N_{Az} = -\mathcal{D}_{AB}(\partial c_A/\partial z) + x_A(N_{Az} + N_{Bz})$, as given in many engineering texts.)

Solution. a. A mass balance on a spherical shell leads to

$$\frac{d}{dr}(r^2 N_{Ar}) = 0 \quad (17.2-18)$$

Substitution of the expression for molar flux N_{Ar} from Eq. 16.2-2 with $N_{Br} = 0$ then gives

$$\frac{d}{dr} \left(r^2 \frac{c \mathcal{D}_{AB}}{1 - x_A} \frac{dx_A}{dr} \right) = 0 \quad (17.2-19)$$

For constant temperature the product $c \mathcal{D}_{AB}$ is constant, and Eq. 17.2-19 may be integrated to give the concentration distribution

$$\left(\frac{x_B}{x_{B1}} \right) = \left(\frac{x_{B2}}{x_{B1}} \right)^{\frac{(1/r_1) - (1/r)}{(1/r_1) - (1/r_2)}} \quad (17.2-20)$$

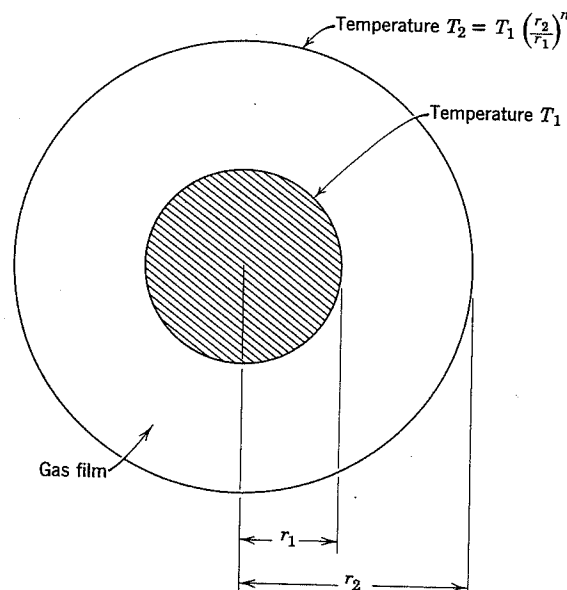


Fig. 17.2-3. Diffusion through a spherical film.

with $x_A = 1 - x_B$. From Eq. 17.2-20 one can then get

$$\mathcal{W}_A = 4\pi r_1^2 N_{Ar}|_{r=r_1} = \frac{4\pi c \mathcal{D}_{AB}}{(1/r_1) - (1/r_2)} \ln \frac{x_{B2}}{x_{B1}} \quad (17.2-21)$$

which is the molar flow of A across any spherical surface.

b. For the nonisothermal situation, combination of Eqs. 17.2-16 and 17 gives the variation of diffusivity with position:

$$\left(\frac{\mathcal{D}_{AB}}{\mathcal{D}_{AB,1}} \right) = \left(\frac{r}{r_1} \right)^{3n/2} \quad (17.2-22)$$

When this expression is inserted into Eq. 17.2-19 and c is set equal to p/RT , we get

$$\frac{d}{dr} \left(r^2 \frac{p \mathcal{D}_{AB,1}/RT_1}{1 - x_A} \left(\frac{r}{r_1} \right)^{n/2} \frac{dx_A}{dr} \right) = 0 \quad (17.2-23)$$

When this equation is integrated between $r = r_1$ and $r = r_2$, we can obtain (for $n \neq -2$)

$$\mathcal{W}_A = 4\pi r_1^2 N_{Ar}|_{r=r_1} = \frac{4\pi(p \mathcal{D}_{AB,1}/RT_1)[1 + (n/2)]}{(r_1^{-1-(n/2)} - r_2^{-1-(n/2)})r_1^{n/2}} \ln \frac{x_{B2}}{x_{B1}} \quad (17.2-24)$$

For $n = 0$, this result simplifies to that in Eq. 17.2-21.

§17.3 DIFFUSION WITH HETEROGENEOUS CHEMICAL REACTION

Consider a catalytic reactor, such as that shown in Fig. 17.3-1a, in which the dimerization reaction $2A \rightarrow A_2$ is being carried out. A system of this

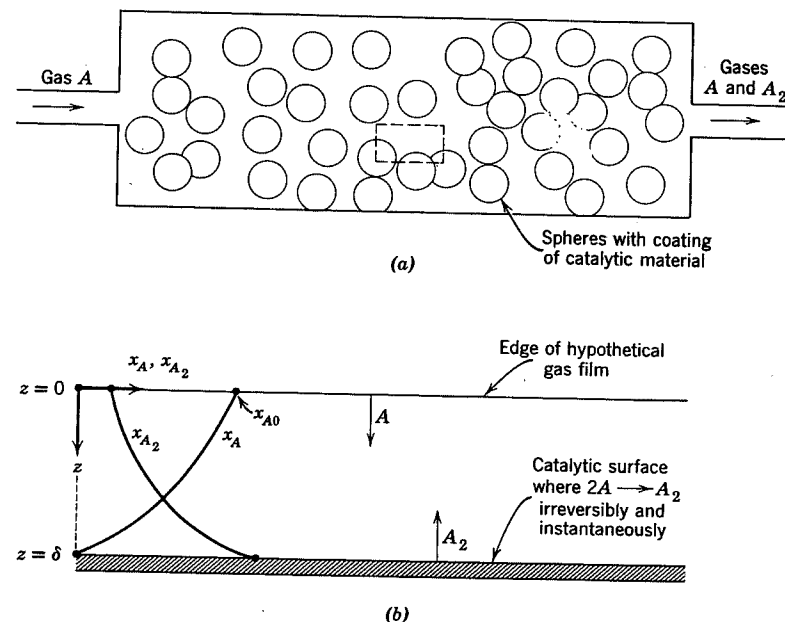


Fig. 17.3-1. (a) Schematic diagram of a catalytic reactor in which A is being converted into A_2 . (b) Idealized picture (or "model") of the diffusion problem near a catalyst particle—that is, the enlargement of dotted region in (a).

degree of complexity cannot be described exactly by a theoretical development. Some information about the behavior of the system can, however, be obtained by analysis of a highly simplified model.

For example, we may imagine that each catalyst particle is surrounded by a stagnant gas film through which A has to diffuse in order to arrive at the catalytic surface. (See Fig. 17.3-1b.) At the catalyst surface we presume that the reaction $2A \rightarrow A_2$ occurs instantaneously and that the product A_2 then diffuses back out through the gas film to the main turbulent gas stream

composed of A and A_2 . We seek then an expression for the local rate of conversion from A to A_2 when the effective gas-film thickness δ and the main gas-stream compositions x_{A0} and x_{A20} are known. We assume that the gas film is isothermal, although in many catalytic reactions the heat generated by the reaction cannot be neglected.

For the situation depicted in Fig. 17.3-1b, there is *one* mole of A_2 moving in the *minus* z -direction for every *two* moles of A moving in the *plus* z -direction. We know this from the stoichiometry of the reaction. Hence we know that at steady state

$$N_{A_2z} = -\frac{1}{2}N_{Az} \quad (17.3-1)$$

at any value of z . This relation may be substituted into Eq. 17.0-1, which may be solved for N_{Az} to give

$$N_{Az} = -\frac{c\mathcal{D}_{AA_2}}{1 - \frac{1}{2}x_A} \frac{dx_A}{dz} \quad (17.3-2)$$

Hence the law of diffusion plus the stoichiometry of the reaction have led us to an expression for N_{Az} in terms of the concentration gradient.

We now make a mass balance on species A over a thin slab of the gas film of thickness Δz . This procedure is exactly the same as that used in connection with Eqs. 17.2-2 and 3 and leads us once again to the equation

$$\frac{dN_{Az}}{dz} = 0 \quad (17.3-3)$$

Insertion of the expression for N_{Az} , developed above, into this equation gives (for constant $c\mathcal{D}_{AA_2}$)

$$\frac{d}{dz} \left(\frac{1}{1 - \frac{1}{2}x_A} \frac{dx_A}{dz} \right) = 0 \quad (17.3-4)$$

Integration twice with respect to z gives

$$-2 \ln \left(1 - \frac{1}{2}x_A \right) = C_1 z + C_2 \quad (17.3-5)$$

The constants of integration are then evaluated by using the boundary conditions

$$\text{B. C. 1:} \quad \text{at } z = 0, \quad x_A = x_{A0} \quad (17.3-6)$$

$$\text{B. C. 2:} \quad \text{at } z = \delta, \quad x_A = 0 \quad (17.3-7)$$

The final result, after a bit of algebraic juggling, is

$$\left(1 - \frac{1}{2}x_A \right) = \left(1 - \frac{1}{2}x_{A0} \right)^{1 - (z/\delta)} \quad (17.3-8)$$

for the concentration profile in the gas film. Equation 17.3-2 may now be used to get the molar flux through the film:

$$N_{Az} = \frac{2c\mathcal{D}_{AA_2}}{\delta} \ln \left(\frac{1}{1 - \frac{1}{2}x_{A0}} \right) \quad (17.3-9)$$

The quantity N_{Az} may also be interpreted as the local rate of dimerization per unit area of catalytic surface. This information can be combined with other information about the catalytic reactor sketched in Fig. 17.3-1a to obtain information about the over-all conversion rate in the entire reactor.

One point deserves to be emphasized. Although the chemical reaction occurs instantaneously at the catalytic surface, the conversion of A to A_2 proceeds at a finite rate because of the diffusion process which is "in series" with the reaction process. Hence we speak of the conversion of A to A_2 as being *diffusion-controlled*.

In the foregoing discussion we have assumed that the dimerization occurs instantaneously at the catalytic surface. In the illustrative example that follows we show how one can account for finite reaction rates at the catalytic surface.

Example 17.3-1. Diffusion with Slow Heterogeneous Reaction

Rework the problem just considered in §17.3 when the reaction $2A \rightarrow A_2$ is not instantaneous at the catalytic surface (at $z = \delta$). Instead, assume that the rate at which A disappears at the catalyst-coated surface is proportional to the concentration of A at that surface:

$$N_{Az} = k_1'' c_A = ck_1'' x_A \quad (17.3-10)$$

in which k_1'' is a rate constant.

Solution. We proceed exactly as before, except that B. C. 2 in Eq. 17.3-7 must be replaced by

$$\text{B. C. 2':} \quad \text{at } z = \delta, \quad x_A = \frac{N_{Az}}{ck_1''} \quad (17.3-11)$$

N_{Az} being, of course, a constant at steady state. The determination of the integration constants for B. C. 1 and B. C. 2' leads to

$$\left(1 - \frac{1}{2}x_A \right) = \left(1 - \frac{1}{2} \frac{N_{Az}}{ck_1''} \right)^{z/\delta} \left(1 - \frac{1}{2}x_{A0} \right)^{1 - (z/\delta)} \quad (17.3-12)$$

Then we may calculate N_{Az} from Eq. 17.3-2 at $z = 0$ and get

$$N_{Az} = \frac{2c\mathcal{D}_{AA_2}}{\delta} \ln \left(\frac{1 - \frac{1}{2}(N_{Az}/ck_1'')}{1 - \frac{1}{2}x_{A0}} \right) \quad (17.3-13)$$

This is a transcendental equation for N_{Az} as a function of x_{A0} , k_1'' , $c\mathcal{D}_{AA_2}$, and δ . When k_1'' is large, the logarithm of $1 - \frac{1}{2}(N_{Az}/ck_1'')$ may be expanded in a Taylor series and all terms discarded but the first. We then get

$$N_{Az} = \frac{2c\mathcal{D}_{AA_2}/\delta}{1 + (\mathcal{D}_{AA_2}/k_1''\delta)} \ln \left(\frac{1}{1 - \frac{1}{2}x_{A0}} \right) \quad (17.3-14)$$

Note once again that the rate of the *combined* reaction and diffusion process is obtained by the foregoing calculation. Note also that the dimensionless group $\mathcal{D}_{AA_2}/k_1''\delta$ describes the effect of the surface reaction rate on the over-all diffusion-reaction process.

§17.4 DIFFUSION WITH HOMOGENEOUS CHEMICAL REACTION

As the next illustration of setting up a mass balance, we consider the system shown in Fig. 17.4-1. Here gas A dissolves in liquid B and diffuses into the liquid phase. As it diffuses, A also undergoes an irreversible first-order chemical reaction: $A + B \rightarrow AB$. Hence the mass balance (input - output + production = 0) takes the form

$$N_{Az}|_z - N_{Az}|_{z+\Delta z} S - k_1''' c_A S \Delta z = 0 \quad (17.4-1)$$

in which k_1''' is a first-order rate constant for the chemical decomposition of

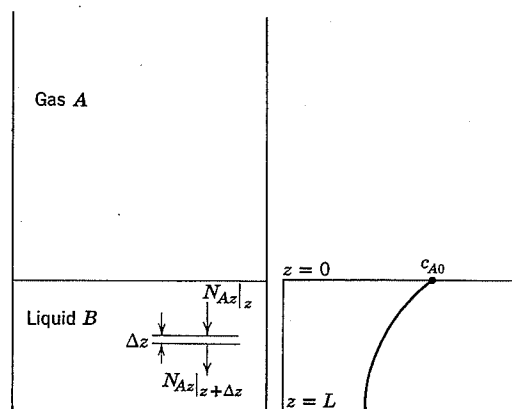


Fig. 17.4-1. Absorption with a homogeneous chemical reaction.

A , and S is the cross-sectional area of the liquid. The quantity $k_1''' c_A$ represents the moles of A disappearing per unit volume per unit time. Division of Eq. 17.4-1 by $S \Delta z$ and taking the limit as Δz goes to zero gives

$$\frac{dN_{Az}}{dz} + k_1''' c_A = 0 \quad (17.4-2)$$

If A and AB are present in small concentrations, then we may to a good approximation write Eq. 17.0-1 as

$$N_{Az} = -\mathcal{D}_{AB} \frac{dc_A}{dz} \quad (17.4-3)$$

Substitution of Eq. 17.4-3 into 17.4-2 gives

$$-\mathcal{D}_{AB} \frac{d^2 c_A}{dz^2} + k_1''' c_A = 0 \quad (17.4-4)$$

This is to be solved with the boundary conditions

$$\text{B. C. 1:} \quad \text{at } z = 0, \quad c_A = c_{A0} \quad (17.4-5)$$

$$\text{B. C. 2:} \quad \text{at } z = L, \quad N_{Az} = 0 \quad \text{or} \quad \frac{dc_A}{dz} = 0 \quad (17.4-6)$$

The first boundary condition states that the surface concentration is maintained at a fixed value c_{A0} . The second states that no A diffuses through the bottom of the container. The solution of Eq. 17.4-4, subject to these boundary conditions, gives

$$\frac{c_A}{c_{A0}} = \frac{\cosh b_1 [1 - (z/L)]}{\cosh b_1} \quad (17.4-7)$$

in which $b_1 = \sqrt{k_1''' L^2 / \mathcal{D}_{AB}}$. The concentration profile thus obtained is shown in Fig. 17.4-1.

From Eq. 17.4-7 one can obtain the average concentration of A in the liquid phase:

$$\begin{aligned} \frac{c_{A, \text{avg}}}{c_{A0}} &= \frac{\int_0^L (c_A/c_{A0}) dz}{\int_0^L dz} \\ &= \frac{\int_0^1 (\cosh b_1 \zeta) / (\cosh b_1) d\zeta}{\int_0^1 d\zeta} \\ &= \frac{\sinh b_1 \zeta}{b_1 \cosh b_1} \Big|_0^1 = \frac{1}{b_1} \tanh b_1 \end{aligned} \quad (17.4-8)$$

In the foregoing ζ is used as an abbreviation for $[1 - (z/L)]$.

The molar flux of A at plane $z = 0$ can be computed from Eq. 17.4-7:

$$\begin{aligned} N_{Az}|_{z=0} &= -\mathcal{D}_{AB} \frac{dc_A}{dz} \Big|_{z=0} \\ &= -\mathcal{D}_{AB} c_{A0} \frac{\sinh b_1 [1 - (z/L)]}{\cosh b_1} \left(-\frac{b_1}{L} \right) \Big|_{z=0} \\ &= + \left(\frac{\mathcal{D}_{AB} c_{A0}}{L} \right) b_1 \tanh b_1 \end{aligned} \quad (17.4-9)$$

in which $b_1 = \sqrt{k_1''' L^2 / \mathcal{D}_{AB}}$.

In this treatment it has been assumed that A is present in small concentration (see Eq. 17.4-3) and that the product of the chemical reaction AB does not interfere with the diffusion of A through B .

Example 17.4-1. Gas Absorption with Chemical Reaction in an Agitated Tank¹

Estimate the effect of chemical reaction rate on the rate of gas absorption in an agitated tank. (See Fig. 17.4-2.) Consider the case in which the dissolved gas A undergoes an irreversible first-order reaction with the liquid B —that is, A disappears in the liquid phase at a rate proportional to the local concentration of A .

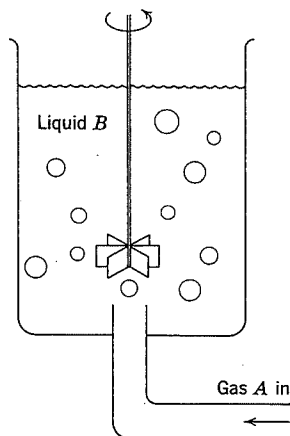


Fig. 17.4-2. Gas-absorption apparatus.

Solution. An exact analysis of this situation is not yet possible because of the complexity of the gas-absorption process. But a useful semiquantitative understanding can be obtained by the analysis of a relatively simple model. The model we employ embodies the following assumptions:

- Each gas bubble is surrounded by a stagnant liquid film of thickness δ , which is small with respect to the bubble diameter.
- A steady-state concentration is quickly established in the liquid film after the bubble is formed.
- The gas A is only sparingly soluble in the liquid, so that we can neglect the bulk flow term in Eq. 17.0-1.
- The bulk of the liquid outside the stagnant film is at concentration $c_{A\delta}$, which changes so slowly with respect to time that it can be considered to be constant.

The differential equation describing the diffusion with chemical reaction is the same as that in Eq. 17.4-4, but the boundary conditions are now

$$\text{B.C. 1:} \quad \text{at } z = 0, \quad c_A = c_{A0} \quad (17.4-10)$$

$$\text{B.C. 2:} \quad \text{at } z = \delta, \quad c_A = c_{A\delta} \quad (17.4-11)$$

The concentration c_{A0} is the interfacial concentration of A in the liquid phase,

¹ E. N. Lightfoot, *A.I.Ch.E. Journal*, 4, 499-500 (1958).

which is assumed to be at equilibrium with the gas phase at the interface, and $c_{A\delta}$ is the concentration of A in the main body of the liquid. The solution of Eq. 17.4-4 with these boundary conditions is

$$\left(\frac{c_A}{c_{A0}}\right) = \frac{\Gamma \sinh b_1 \zeta + \sinh b_1 (1 - \zeta)}{\sinh b_1} \quad (17.4-12)$$

in which $\zeta = z/\delta$, $\Gamma = c_{A\delta}/c_{A0}$, and $b_1 = \sqrt{k_1'' \delta^2 / \mathcal{D}_{AB}}$. The analogous solution

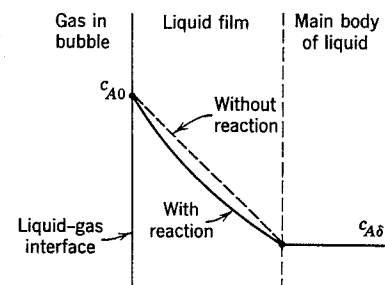


Fig. 17.4-3. Predicted concentration profile in liquid film near a bubble.

in the absence of chemical reaction (that is, $b_1 = 0$) is

$$\left(\frac{c_A}{c_{A0}}\right)_{\text{no reaction}} = \Gamma \zeta + (1 - \zeta) \quad (17.4-13)$$

The concentration profiles from Eqs. 17.4-12 and 17.4-13 are shown in Fig. 17.4-3. From these concentration profiles we can then get the mass fluxes for absorption with and without reaction, which are, at $z = 0$,

$$N_{Az}|_{z=0} = \left(-\mathcal{D}_{AB} \frac{dc_A}{dz}\right)_{z=0} = \frac{\mathcal{D}_{AB} c_{A0}}{\delta} \left(\frac{b_1 \cosh b_1 - b_1 \Gamma}{\sinh b_1}\right) \quad (17.4-14)$$

$$(N_{Az})_{\text{no reaction}} = \left(-\mathcal{D}_{AB} \frac{dc_A}{dz}\right)_{z=0, \text{ no reaction}} = \frac{\mathcal{D}_{AB} c_{A0}}{\delta} (1 - \Gamma) \quad (17.4-15)$$

It is common practice to estimate apparent film thicknesses *in the presence of chemical reaction* by using Eq. 17.4-15 and setting $\Gamma = 0$ in it. This is equivalent to assuming a value of unity for the ratio:

$$N^* = \frac{\left(\text{absorption rate with first-order reaction}\right)}{\left(\text{absorption rate with no reaction and with } c_{A\delta} = 0\right)} = \left(\frac{b_1}{\sinh b_1}\right) (\cosh b_1 - \Gamma) \quad (17.4-16)$$

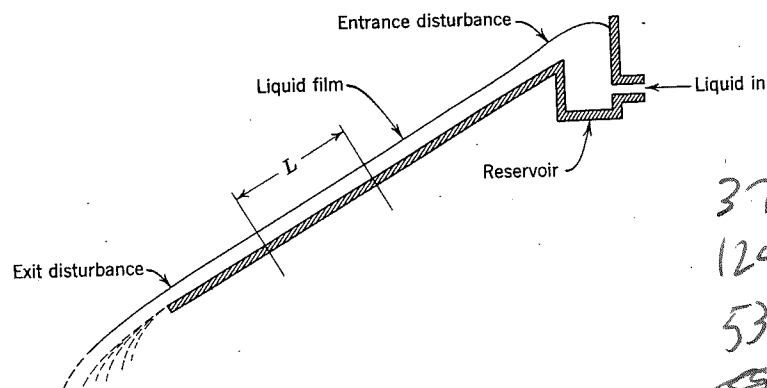


Fig. 2.2-1. Schematic diagram of falling film experiment, illustrating end effects. In the region of length L the velocity distribution is fully developed.

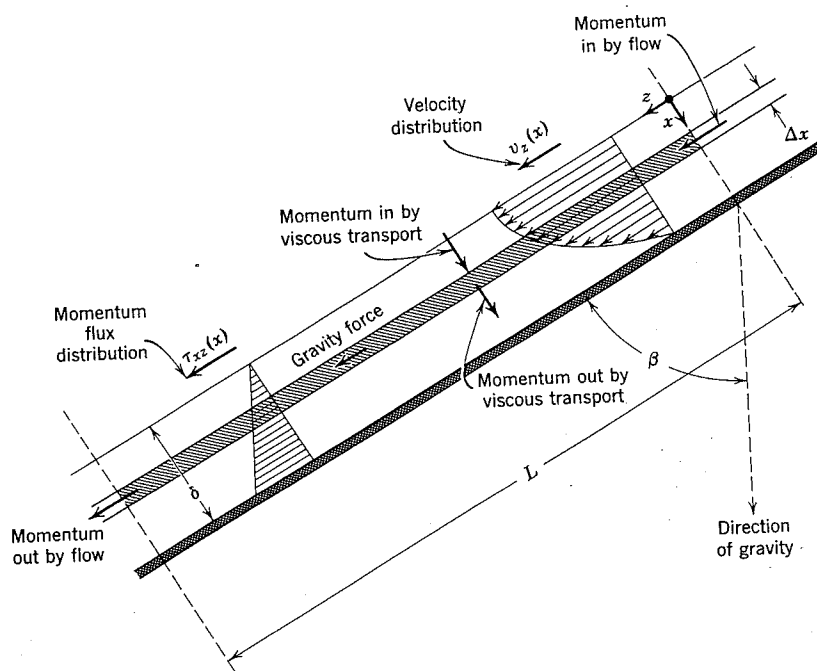


Fig. 2.2-2. Flow of a viscous isothermal liquid film under the influence of gravity with no rippling. Slice of thickness Δx over which momentum balance is made. The y -axis is pointing outward from the plane of paper.

for the velocity distribution. The integration of these two differential equations yields, respectively, the momentum flux and velocity distributions for the system. This information can then be used to calculate various other quantities, such as average velocity, maximum velocity, volume rate of flow, pressure drop, and forces on boundaries.

In the integrations mentioned above, several constants of integration appear, which are evaluated by the use of "boundary conditions," that is, statements of physical facts at specified values of the independent variable. The following are the most used boundary conditions:

a. At solid-fluid interfaces the fluid velocity equals the velocity with which the surface itself is moving; that is, the fluid is assumed to cling to any solid surfaces with which it is in contact.

b. At liquid-gas interfaces the momentum flux (hence the velocity gradient) in the liquid phase is very nearly zero and can be assumed to be zero in most calculations.

c. At liquid-liquid interfaces the momentum flux perpendicular to the interface, and the velocity, are continuous across the interface. (In the notation of §A.5, v and $np + [n \cdot \tau]$ are continuous for planar interfaces).

All three types of boundary conditions are encountered in the sections that follow.

In this section we have endeavored to present some general rules for solving elementary viscous flow problems. We now proceed to illustrate the application of these rules to a number of simple flow systems.

§2.2 FLOW OF A FALLING FILM

As our first example, we consider the flow of a fluid along an inclined flat surface, as shown in Fig. 2.2-1. Such films have been studied in connection with wetted-wall towers, evaporation and gas absorption experiments, and application of coatings to paper rolls. We consider the viscosity and density of the fluid to be constant. We focus our attention on a region of length L , sufficiently far from the ends of the wall that the entrance and exit disturbances are not included in L , that is, in this region the velocity component v_z does not depend on z .

We begin by setting up a z -momentum balance over a system of thickness Δx , bounded by the planes $z = 0$ and $z = L$, and extending a distance W in the y -direction. (See Fig. 2.2-2.) The various components of the momentum balance are then

rate of z -
momentum in
across surface
at x

$$(LW)(\tau_{xz})|_x$$

(2.2-1)

rate of z -
momentum out
across surface
at $x + \Delta x$

$$(LW)(\tau_{xz})|_{x+\Delta x} \quad (2.2-2)$$

rate of z -
momentum in
across surface
at $z = 0$

$$(W\Delta x v_z)(\rho v_z)|_{z=0} \quad (2.2-3)$$

rate of z -
momentum out
across surface
at $z = L$

$$(W\Delta x v_z)(\rho v_z)|_{z=L} \quad (2.2-4)$$

gravity force
acting on fluid

$$(LW\Delta x)(\rho g \cos \beta) \quad (2.2-5)$$

Note that we always take the "in" and "out" directions in the direction of the positive x - and z -axes (in this problem these happen to coincide with the direction of momentum transport). The notation $|_{x+\Delta x}$ means "evaluated at $x + \Delta x$."

When these terms are substituted into the momentum balance of Eq. 2.1-1, we get

$$LW\tau_{xz}|_x - LW\tau_{xz}|_{x+\Delta x} + W\Delta x \rho v_z^2|_{z=0} - W\Delta x \rho v_z^2|_{z=L} + LW\Delta x \rho g \cos \beta = 0 \quad (2.2-6)$$

Because v_z is the same at $z = 0$ as it is at $z = L$ for each value of x , the third and fourth terms just cancel one another. We now divide Eq. 2.2-6 by $LW\Delta x$ and take the limit as Δx approaches zero:

$$\lim_{\Delta x \rightarrow 0} \left(\frac{\tau_{xz}|_{x+\Delta x} - \tau_{xz}|_x}{\Delta x} \right) = \rho g \cos \beta \quad (2.2-7)$$

The quantity on the left side may be recognized as the definition of the first derivative of τ_{xz} with respect to x . Therefore, Eq. 2.2-7 may be rewritten as

$$\frac{d}{dx} \tau_{xz} = \rho g \cos \beta \quad (2.2-8)$$

This is the differential equation for the momentum flux τ_{xz} . It may be integrated to give

$$\tau_{xz} = \rho g x \cos \beta + C_1 \quad (2.2-9)$$

The constant of integration may be evaluated by making use of the boundary condition at the liquid-gas interface (see §2.1):

$$\text{B.C. 1:} \quad \text{at } x = 0, \quad \tau_{xz} = 0 \quad (2.2-10)$$

Substitution of this boundary condition into Eq. 2.2-9 reveals that $C_1 = 0$. Hence the momentum-flux distribution is

$$\tau_{xz} = \rho g x \cos \beta \quad (2.2-11)$$

as shown in Fig. 2.2-2.

If the fluid is Newtonian, then we know that the momentum flux is related to the velocity gradient according to

$$\tau_{xz} = -\mu \frac{dv_z}{dx} \quad (2.2-12)$$

Substitution of this expression for τ_{xz} into Eq. 2.2-11 gives the following differential equation for the velocity distribution:

$$\frac{dv_z}{dx} = -\left(\frac{\rho g \cos \beta}{\mu} \right) x \quad (2.2-13)$$

This equation is easily integrated to give

$$v_z = -\left(\frac{\rho g \cos \beta}{2\mu} \right) x^2 + C_2 \quad (2.2-14)$$

The constant of integration is evaluated by using the boundary condition

$$\text{B.C. 2:} \quad \text{at } x = \delta, \quad v_z = 0 \quad (2.2-15)$$

Substitution of this boundary condition into Eq. 2.2-14 shows that $C_2 = (\rho g \cos \beta / 2\mu) \delta^2$. Therefore, the velocity distribution is

$$v_z = \frac{\rho g \delta^2 \cos \beta}{2\mu} \left[1 - \left(\frac{x}{\delta} \right)^2 \right] \quad (2.2-16)$$

Hence the velocity profile is parabolic. (See Fig. 2.2-2.)

Once the velocity profile has been found, a number of quantities may be calculated:

(i) The *maximum velocity* $v_{z,\max}$ is clearly the velocity at $x = 0$; that is

$$v_{z,\max} = \frac{\rho g \delta^2 \cos \beta}{2\mu} \quad (2.2-17)$$

(ii) The *average velocity* $\langle v_z \rangle$ over a cross section of the film is obtained by the following calculation:

$$\begin{aligned}\langle v_z \rangle &= \frac{\int_0^W \int_0^\delta v_z dx dy}{\int_0^W \int_0^\delta dx dy} \\ &= \frac{1}{\delta} \int_0^\delta v_z dx \\ &= \frac{\rho g \delta^2 \cos \beta}{2\mu} \int_0^1 \left[1 - \left(\frac{x}{\delta} \right)^2 \right] d\left(\frac{x}{\delta} \right) \\ &= \frac{\rho g \delta^2 \cos \beta}{3\mu} \quad (2.2-18)\end{aligned}$$

(iii) The *volume rate of flow* Q is obtained from the average velocity or by integration of the velocity distribution:

$$Q = \int_0^W \int_0^\delta v_z dx dy = W \delta \langle v_z \rangle = \frac{\rho g W \delta^3 \cos \beta}{3\mu} \quad (2.2-19)$$

(iv) The *film thickness* δ may be given in terms of the average velocity, the volume rate of flow, or the mass rate of flow per unit width of wall ($\Gamma = \rho \delta \langle v_z \rangle$):

$$\delta = \sqrt{\frac{3\mu \langle v_z \rangle}{\rho g \cos \beta}} = \sqrt[3]{\frac{3\mu Q}{\rho g W \cos \beta}} = \sqrt[3]{\frac{3\mu \Gamma}{\rho^2 g \cos \beta}} \quad (2.2-20)$$

(v) The *z-component of the force* F of the fluid on the surface is given by integrating the momentum flux over the fluid-solid interface:

$$\begin{aligned}F_z &= \int_0^L \int_0^W \tau_{xz}|_{x=\delta} dy dz \\ &= \int_0^L \int_0^W -\mu \frac{dv_z}{dx} \Big|_{x=\delta} dy dz \\ &= (LW)(-\mu) \left(-\frac{\rho g \delta \cos \beta}{\mu} \right) \\ &= \rho g \delta LW \cos \beta \quad (2.2-21)\end{aligned}$$

This is clearly just the *z-component of the weight* of the entire fluid in the film.

The foregoing analytical results are valid only when the film is falling in laminar flow with straight streamlines. For the slow flow of thin viscous films, these conditions are satisfied. It has been found experimentally that as the film velocity $\langle v_z \rangle$ increases, as the thickness of the film δ increases, and as

the kinematic viscosity $\nu = \mu/\rho$ decreases, the nature of the flow gradually changes; in this gradual change three more or less distinct stable types of flow can be observed: (a) laminar flow with straight streamlines, (b) laminar flow with rippling, and (c) turbulent flow. Quantitative information concerning the type of flow that can be expected under a given set of physical conditions seems to be only fragmentary. For vertical walls, the following information may be given:^{1,2}

laminar flow without rippling	$\text{Re} < 4$ to 25
laminar flow with rippling	4 to $25 < \text{Re} < 1000$ to 2000
turbulent flow	$\text{Re} > 1000$ to 2000

in which $\text{Re} = 4 \delta \langle v_z \rangle \rho / \mu = 4\Gamma/\mu$ is the *Reynolds number* for this system. Why this dimensionless group is used as a criterion for flow patterns is discussed further in subsequent chapters.

Example 2.2-1. Calculation of Film Velocity

An oil has a kinematic viscosity of $2 \times 10^{-4} \text{ m}^2 \text{ sec}^{-1}$ and a density of $0.8 \times 10^3 \text{ kg m}^{-3}$. What should the mass rate of flow of this film down a vertical wall be in order to have a film thickness of 2.5 mm?

Solution. According to Eq. 2.2-20, the mass rate of flow per unit width of wall is (all numerical values given in mks units)

$$\begin{aligned}\Gamma &= \frac{\delta^3 \rho g}{3\nu} = \frac{(2.5 \times 10^{-3})^3 (0.8 \times 10^3) (9.80)}{3(2 \times 10^{-4})} \\ &= 0.204 \text{ kg m}^{-1} \text{ sec}^{-1}\end{aligned}$$

This is the desired result if and only if the flow is indeed laminar. To ascertain the nature of the flow, we calculate a Reynolds number based on the mass rate of flow just found:

$$\text{Re} = \frac{4\Gamma}{\mu} = \frac{4\Gamma}{\rho\nu} = \frac{4(0.204)}{(0.8 \times 10^3)(2 \times 10^{-4})} = 5.1 \quad (\text{dimensionless})$$

This Reynolds number is below the observed upper limit for laminar flow stated above, and therefore the calculated value of Γ is valid.

Example 2.2-2. Falling Film with Variable Viscosity

Rework the falling film problem for the situation in which the viscosity depends upon position in the following manner:

$$\mu = \mu_0 e^{-\alpha(x/\delta)} \quad (2.2-22)$$

¹ T. K. Sherwood and R. L. Pigford, *Absorption and Extraction*, McGraw-Hill, New York (1952), p. 265.

² S. S. Grimley, *Trans. Inst. Chem. Engrs.* (London), 23, 228-235 (1948).

b. The inertial term $[\mathbf{v} \cdot \nabla \mathbf{v}]$ is either identically zero or negligibly small, as for "creeping flow."

c. The external force $\mathbf{g}$ is derivable from a potential Φ (that is, $\mathbf{g} = -\nabla\Phi$).

d. The velocity $\mathbf{v}$ is known to be $\mathbf{v}_0(x, y, z)$ on the surface S of the flow system of volume V .

For these flows, the integral

$$B = \iiint_{\substack{\text{volume of flow} \\ \text{system with fixed} \\ \text{boundaries}}} F \, dx \, dy \, dz \quad (3.X-1)$$

is a minimum, in which

$$F = -(p + \rho\Phi)I_1 + \frac{1}{2} \int_0^{I_2} \eta(I_2) \, dI_2 \quad (3.X-2)$$

Here I_1 and I_2 are the first and second invariants of Δ . (See Eqs. 3.6-3 and 4.) That is, of all the functions $\mathbf{v}(x, y, z)$ which reduce to the given function $\mathbf{v}_0(x, y, z)$ on S , the true velocity distribution will be that which gives the minimum value of B . In using the foregoing variational principle, one would normally select a trial function $\mathbf{v}(x, y, z)$ such that the equation of continuity is automatically satisfied. Then I_1 computed from the trial velocity distribution vanishes, and it is not necessary to select a trial function $p(x, y, z)$ for the pressure.

a. Show that the Euler-Lagrange equations⁴ corresponding to the foregoing variational principle are just the equation of continuity and the three components of the equation of motion.

b. Verify that this variational principle simplifies to Helmholtz's principle⁵ for a Newtonian fluid.

3.Y₄ Radial Flow between Parallel Disks

In solving the equations of motion one usually postulates the form of the solution for the velocity profiles. These postulates are then used to discard many terms in the equations so that one finally obtains simple starting equations for the problem at hand. Such a procedure is clearly intuitive in nature and can occasionally lead to difficulties.

To illustrate this point, consider the radial flow between a pair of circular disks. In solving this problem (see Problem 3.K) it was *postulated* that $v_r = v_r(r, z)$, and that v_θ and v_z are both zero, and that $p = p(r)$. These postulates lead to Eq. 3.K-1.

Differentiate this equation with respect to z and show that it leads to an inconsistency, except in the limit of creeping flow. Discuss.

⁴ See, for example, E. Madelung, *Die mathematischen Hilfsmittel des Physikers*, Dover, New York (1943), p. 216, Eq. 4.

⁵ H. Lamb, *Hydrodynamics*, Dover, New York (1945), pp. 617-619; see also D. F. Hays, *Journal of Basic Engineering*, **81**, 13-23 (1959) for an interesting application to lubrication problems.

Velocity Distributions with More Than One Independent Variable

In Chapter 2 it was shown that a variety of steady-state viscous flow problems with straight streamlines may be solved by shell momentum balances. In Chapter 3 more general methods were developed, which enable us to describe more complicated flow systems. In §3.5 a few applications of the general equations were given, but in order to emphasize the physics and minimize the mathematics these applications were restricted for the most part to problems leading to ordinary rather than to partial differential equations. In this chapter we discuss several classes of applications that involve the solutions of partial differential equations: unsteady flow, viscous flow in more than one direction, two-dimensional ideal flow, and viscous flow in boundary layers. The purpose of this chapter is to provide an introduction to each of these topics along with some of the standard mathematical techniques. None of these topics is discussed in detail, inasmuch as all are treated extensively in fluid mechanics treatises.

§4.1 UNSTEADY VISCOUS FLOW

The study of transient phenomena of all types is becoming important in connection with current interest in problems of start-up and control. The

equations given in Chapter 3 provide the starting point for describing unsteady flow. We give here two illustrations of how such problems are set up and solved.¹

In the first we examine the flow of an infinite body of fluid near a wall suddenly set in motion. This problem will illustrate the use of the *method of combination of variables*, which enables one to reduce the partial differential equation to a single ordinary differential equation; such a procedure is possible only when two of the boundary conditions can be combined into one. It should be noted that this system never approaches a limiting steady state.

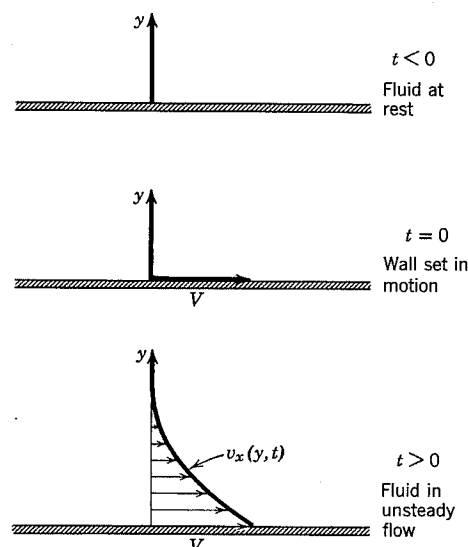


Fig. 4.1-1. Viscous flow of a fluid near a wall suddenly set in motion.

In the second illustration we consider the transient behavior of a viscous fluid in a tube. This problem is handled by the *method of separation of variables*, which enables one to reduce the partial differential equation to two ordinary differential equations. This system does have a limiting velocity profile as time goes to infinity—that is, the system approaches a steady state.

Example 4.1-1. Flow Near a Wall Suddenly Set in Motion

A semi-infinite body of liquid with constant ρ and μ is bounded on one side by a flat surface (the xz -plane). Initially, the fluid and the solid surface are at rest; but at time $t = 0$ the solid surface is set in motion in the positive x -direction with

¹ Other examples may be found in H. Lamb, *Hydrodynamics*, Dover, New York (1945), Chapter 11.

a velocity V , as shown in Fig. 4.1-1. It is desired to know the velocity as a function of y and t . There is no pressure gradient or gravity force in the x -direction, and the flow is assumed to be laminar.

Solution. For this system, $v_y = v_z = 0$ and $v_x = v_x(y, t)$. Then from Eq. A of Table 3.4-1 (continuity) and Eq. D of Table 3.4-2 (motion) we find that

$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2} \quad (4.1-1)$$

in which $\nu = \mu/\rho$. The initial and boundary conditions are

$$\text{I.C.:} \quad \text{at } t \leq 0, \quad v_x = 0 \quad \text{for all } y \quad (4.1-2)$$

$$\text{B.C. 1:} \quad \text{at } y = 0, \quad v_x = V \quad \text{for all } t > 0 \quad (4.1-3)$$

$$\text{B.C. 2:} \quad \text{at } y = \infty, \quad v_x = 0 \quad \text{for all } t > 0 \quad (4.1-4)$$

We now anticipate that a solution can be found of the form $v_x/V = \phi(\eta)$, in which η is a dimensionless variable $y/\sqrt{4\nu t}$. In terms of the new variables,

$$\frac{\partial(v_x/V)}{\partial t} = -\frac{1}{2} \frac{\eta}{t} \phi'; \quad \frac{\partial^2(v_x/V)}{\partial y^2} = + \frac{\eta^2}{y^2} \phi'' \quad (4.1-5,6)$$

in which primes indicate differentiation with respect to η . Then Eq. 4.1-1 becomes

$$\phi'' + 2\eta\phi' = 0 \quad (4.1-7)$$

with the boundary conditions

$$\text{B.C. 1:} \quad \text{at } \eta = 0, \quad \phi = 1 \quad (4.1-8)$$

$$\text{I.C. + B.C. 2:} \quad \text{at } \eta = \infty, \quad \phi = 0 \quad (4.1-9)$$

If ϕ' is replaced by ψ , then we get a first-order separable equation, which may be solved to give

$$\psi \equiv \phi' = C_1 e^{-\eta^2} \quad (4.1-10)$$

A second integration then gives

$$\phi = C_1 \int_0^\eta e^{-\eta^2} d\eta + C_2 \quad (4.1-11)$$

where we have arbitrarily selected $\eta = 0$ for the lower limit of the indefinite integral, which we cannot evaluate in closed form; changing the lower limit $\eta = 0$ to a different limit would simply change the value of constant C_2 , still undetermined. Application of the two boundary conditions then gives

$$\phi = 1 - \frac{\int_0^\eta e^{-\eta^2} d\eta}{\int_0^\infty e^{-\eta^2} d\eta} = 1 - \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-\eta^2} d\eta \quad (4.1-12)$$

The ratio of the integrals appearing here is called the “error function,” abbreviated $\text{erf } \eta$; it is a well-known function, and tables of it are readily available.² Hence the

² See, for example, H. B. Dwight *Tables of Integrals and Other Mathematical Data*, Macmillan, New York (1957), Third Edition, p. 275; $1 - \text{erf } u$ is sometimes written as $\text{erfc } u$, the “complementary error function.”

solution may be written

$$\frac{v_x}{V} = 1 - \operatorname{erf} \frac{y}{\sqrt{4\nu t}} \quad (4.1-13)$$

The instantaneous velocity profiles are shown in Fig. 4.1-2; we also show the analogous profiles for some power-model non-Newtonian fluids.³

The error function is a monotone increasing function ranging from 0 to 1 and reaches a value of 0.99 when η is about 2.0. We can use this fact to define a "boundary-layer thickness" δ as that distance y for which v_x has dropped to a value $0.01V$. That is, we find that $\delta = 4\sqrt{\nu t}$. This distance is a measure of the extent to which momentum has "penetrated" the body of the fluid. Note that this thickness is proportional to the square root of the time.

Example 4.1-2. Unsteady Laminar Flow in a Circular Tube

A fluid of constant ρ and μ is contained in a very long horizontal pipe of length L and radius R . Initially the fluid is at rest. At $t = 0$, a pressure gradient $(p_0 - p_L)/L$ is impressed on the system. Determine how the velocity profiles change with time.

Solution. For this problem, cylindrical coordinates are used; it is assumed that $v_r = v_\theta = 0$ and that $v_z = v_z(r, t)$. Then Eq. B of Table 3.4-1 (continuity) and Eq. F of Table 3.4-3 (motion) may be combined to give

$$\rho \frac{\partial v_z}{\partial t} = \frac{p_0 - p_L}{L} + \mu \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) \quad (4.1-14)$$

The initial and boundary conditions are

$$\text{I.C.:} \quad \text{at } t = 0, \quad v_z = 0 \quad \text{for } 0 \leq r \leq R \quad (4.1-15)$$

$$\text{B.C. 1:} \quad \text{at } r = 0, \quad v_z = \text{finite} \quad (4.1-16)$$

$$\text{B.C. 2:} \quad \text{at } r = R, \quad v_z = 0 \quad (4.1-17)$$

First we introduce the following dimensionless variables:

$$\phi = \frac{v_z}{(p_0 - p_L)R^2/4\mu L}; \quad \xi = \frac{r}{R}; \quad \tau = \frac{\mu t}{\rho R^2} \quad (4.1-18, 19, 20)$$

The velocity is made dimensionless by dividing by the maximum velocity at $t = \infty$. When Eq. 4.1-14 is multiplied by $4L/(p_0 - p_L)$, we get in dimensionless variables

$$\frac{\partial \phi}{\partial \tau} = 4 + \frac{1}{\xi} \frac{\partial}{\partial \xi} \left(\xi \frac{\partial \phi}{\partial \xi} \right) \quad (4.1-21)$$

This is to be solved with the conditions that at $\tau = 0$, $\phi = 0$, at $\xi = 1$, $\phi = 0$, and at $\xi = 0$, ϕ is finite.

Now we make use of the fact that the system will attain a steady state at $\tau = \infty$. Hence a solution of the following form must be sought:

$$\phi(\xi, \tau) = \phi_\infty(\xi) - \phi_i(\xi, \tau) \quad (4.1-22)$$

³ R. B. Bird, *A.I.Ch.E. Journal*, 5, 565 (1959).

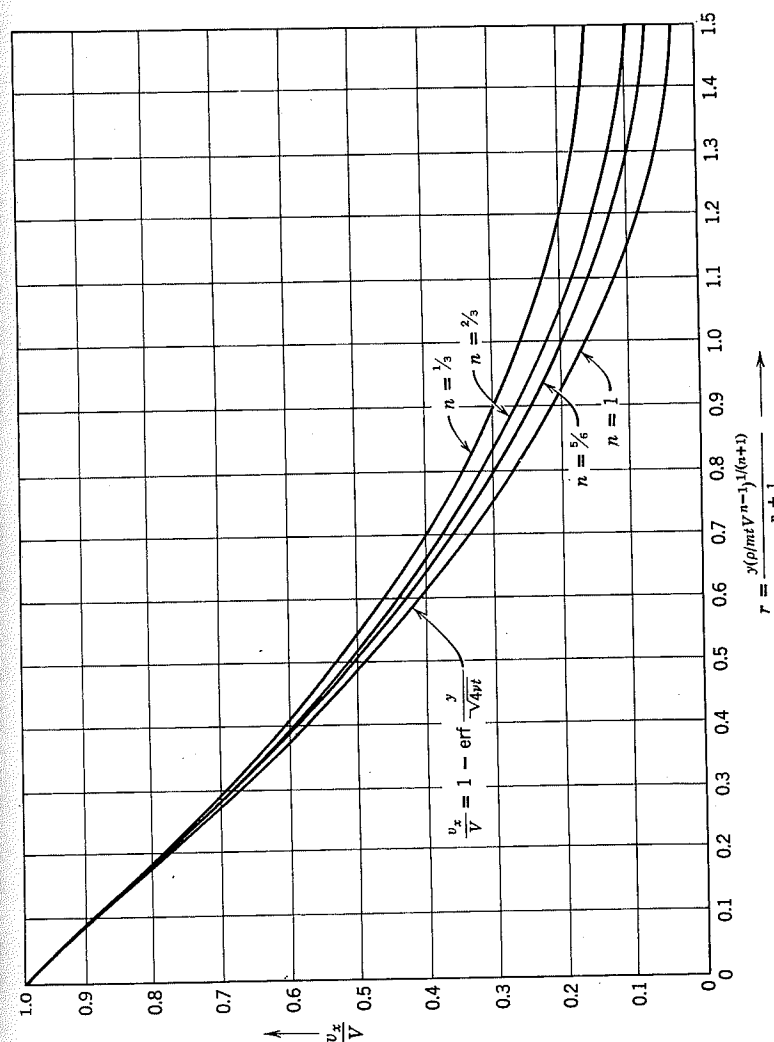


Fig. 4.1-2. Velocity distribution (in dimensionless form) for flow in the neighborhood of a wall suddenly set in motion. The curves are drawn for the power model with parameters m and n . For $n = 1$ (Newtonian flow) the variable r becomes $y/\sqrt{4\nu t}$.

To determine the conditions under which this assumption is valid, we need an expression for Γ . This is obtained by equating the mass of A crossing the outer surface of the film with the amount of A being removed by chemical reaction in the bulk of the liquid [here we make use of assumption (d) above]:

$$-A \mathcal{D}_{AB} \left. \frac{dc_A}{dz} \right|_{z=\delta} = V k_1''' c_{A\delta} \quad (17.4-17)$$

in which A is the total bubble surface area and V is the volume of the main body

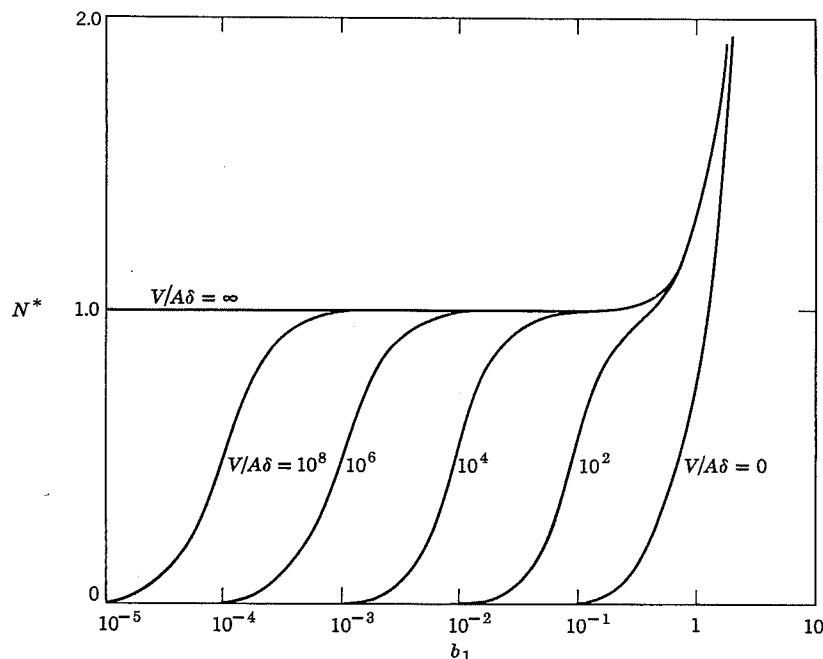


Fig. 17.4-4. Gas absorption accompanied by an irreversible first-order chemical reaction.

of liquid. Using Eq. 17.4-12, we evaluate $dc_A/dz|_{z=\delta}$ and $c_{A\delta}$ to get

$$\Gamma = \frac{1}{\cosh b_1 + b_1(V/A\delta) \sinh b_1} \quad (17.4-18)$$

We can then rewrite Eq. 17.4-16 as

$$N^* = \frac{b_1}{\sinh b_1} \left(\cosh b_1 - \frac{1}{\cosh b_1 + b_1(V/A\delta) \sinh b_1} \right) \quad (17.4-19)$$

This result is plotted in Fig. 17.4-4. It can be seen from this graph that over a

fairly wide range of b_1 the ratio N^* is indeed very nearly unity, and Eq. 17.4-15 may be used to describe the absorption rate merely by setting $\Gamma = 0$. Similar simplification is possible for more realistic mass-transfer models.¹

§17.5 DIFFUSION INTO A FALLING LIQUID FILM: FORCED-CONVECTION MASS TRANSFER

In this section we present an illustration of forced-convection mass transfer in which viscous flow and diffusion occur under such conditions that the

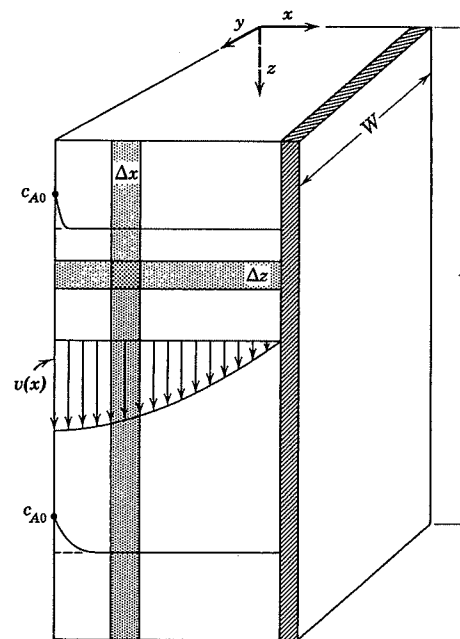


Fig. 17.5-1. Absorption into a falling film.

velocity field can be considered virtually unaffected by the diffusion. Specifically, we consider the absorption of gas A by a laminar falling film of liquid B . The material A is only slightly soluble in B , so that the viscosity of the liquids is not changed appreciably. We shall make the further restriction that the diffusion take place so slowly in the liquid film that A will not "penetrate" very far into B —that the penetration distance be small in comparison with the film thickness. The system is sketched in Fig. 17.5-1.

Let us now set up the differential equations describing the process. First of all we have to solve the momentum transfer problem to obtain the velocity profile $v_z(x)$ for the film; this has already been worked out in §2.2 in the

absence of mass transfer at the fluid surface, and we know that the result is

$$v_z(x) = v_{\max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right] \quad (17.5-1)$$

provided that "end effects" are not considered.

Next we have to establish a mass balance on component A . We note that c_A will be changing both with x and with z . Hence as the element of volume over which we set up the mass balance we select the volume formed by the intersection of a slab of thickness Δz with a slab of thickness Δx . Then the mass balance on A is simply

$$N_{Az}|_z W \Delta x - N_{Az}|_{z+\Delta z} W \Delta x + N_{Ax}|_x W \Delta z - N_{Ax}|_{x+\Delta x} W \Delta z = 0 \quad (17.5-2)$$

in which W is the width of the film. By dividing by $W \Delta x \Delta z$ and performing the usual limiting process as the volume element becomes infinitesimally small, we get

$$\frac{\partial N_{Az}}{\partial z} + \frac{\partial N_{Ax}}{\partial x} = 0 \quad (17.5-3)$$

Into this equation we now have to insert the expressions for N_{Az} and N_{Ax} by making appropriate simplifications of Eq. 17.0-1. For the molar flux in the z -direction, we write, assuming constant c

$$N_{Az} = -\mathcal{D}_{AB} \frac{\partial c_A}{\partial z} + x_A(N_{Az} + N_{Bz}) \approx c_A v_z(x) \quad (17.5-4)$$

That is, A moves in the z -direction primarily because of the flow of the film, the diffusive contribution being negligible. The molar flux in the x -direction will be

$$N_{Ax} = -\mathcal{D}_{AB} \frac{\partial c_A}{\partial x} + x_A(N_{Ax} + N_{Bx}) \approx -\mathcal{D}_{AB} \frac{\partial c_A}{\partial x} \quad (17.5-5)$$

That is, in the x -direction A is transported primarily by diffusion, there being almost no convective transport because of the very slight solubility of A in B . Substitution of these expressions for N_{Az} and N_{Ax} into Eq. 17.5-3 gives

$$v_z \frac{\partial c_A}{\partial z} = \mathcal{D}_{AB} \frac{\partial^2 c_A}{\partial x^2} \quad (17.5-6)$$

which is the differential equation describing $c_A(x, z)$.

The forced-convection mass transfer is then described by Eqs. 17.5-1 and 17.5-6. When these two equations are combined, we get finally

$$v_{\max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right] \frac{\partial c_A}{\partial z} = \mathcal{D}_{AB} \frac{\partial^2 c_A}{\partial x^2} \quad (17.5-7)$$

This partial differential equation is to be solved with the following boundary conditions:

$$\text{B. C. 1:} \quad \text{at } z = 0, \quad c_A = 0 \quad (17.5-8)$$

$$\text{B. C. 2:} \quad \text{at } x = 0, \quad c_A = c_{A0} \quad (17.5-9)$$

$$\text{B. C. 3:} \quad \text{at } x = \delta, \quad \frac{\partial c_A}{\partial x} = 0 \quad (17.5-10)$$

when the fluid film begins as pure B , and the interface concentration in the liquid is taken to be the solubility of A in B , here designated as c_{A0} . This problem has been solved by Pigford,¹ but we do not give that solution here. Instead, we solve to obtain only a limiting expression valid for "short contact times"—that is, small values of $L/v_{\max}$.

If, as is indicated in Fig. 17.5-1, the substance A has penetrated only a short distance into the film, then the A species for the most part has the impression that the film is moving throughout with a velocity equal to $v_{\max}$. Furthermore, if A does not penetrate very far, it does not "feel" the presence of the solid wall at $x = \delta$. Hence, if the film were of infinite thickness moving with the velocity $v_{\max}$, the diffusing material would not know the difference. This physical argument suggests then that we replace Eq. 17.5-7 and its boundary conditions by

$$v_{\max} \frac{\partial c_A}{\partial z} = \mathcal{D}_{AB} \frac{\partial^2 c_A}{\partial x^2} \quad (17.5-11)$$

$$\text{B. C. 1:} \quad \text{at } z = 0, \quad c_A = 0 \quad (17.5-12)$$

$$\text{B. C. 2:} \quad \text{at } x = 0, \quad c_A = c_{A0} \quad (17.5-13)$$

$$\text{B. C. 3:} \quad \text{at } x = \infty, \quad c_A = 0 \quad (17.5-14)$$

The solution to Eq. 17.5-11 with these boundary conditions is²

$$\begin{aligned} \frac{c_A}{c_{A0}} &= 1 - \frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{4\mathcal{D}_{AB}z/v_{\max}}}} e^{-\xi^2} d\xi \\ &= 1 - \operatorname{erf} \frac{x}{\sqrt{4\mathcal{D}_{AB}z/v_{\max}}} \end{aligned}$$

or

$$\frac{c_A}{c_{A0}} = \operatorname{erfc} \frac{x}{\sqrt{4\mathcal{D}_{AB}z/v_{\max}}} \quad (17.5-15)$$

¹ R. L. Pigford, doctoral dissertation, University of Illinois (1941).

² The solution is worked out in detail in Example 4.1-1 by the "method of combination of variables."

Here $\text{erf } y$ is the "error function" defined in §4.1 and $\text{erfc } y = 1 - \text{erf } y$ is the "complementary error function"; both are standard tabulated functions. Once the concentration profiles are known, the total mass-transfer rate may be found by either of the following two methods:

Integration of Mass Flux over Length of Film

The local mass flux at the surface $x = 0$ at a position z down the plate is

$$N_{Ax}(z)|_{x=0} = -\mathcal{D}_{AB} \frac{\partial c_A}{\partial x} \bigg|_{x=0} = c_{A0} \sqrt{\frac{\mathcal{D}_{AB} v_{\max}}{\pi z}} \quad (17.5-16)$$

The total moles of A transferred per unit time from the gas to the liquid film is

$$\begin{aligned} \mathcal{W}_A &= \int_0^W \int_0^L N_{Ax}|_{x=0} dz dy \\ &= W c_{A0} \sqrt{\frac{\mathcal{D}_{AB} v_{\max}}{\pi}} \int_0^L z^{-1/2} dz \\ &= W L c_{A0} \sqrt{\frac{4 \mathcal{D}_{AB} v_{\max}}{\pi L}} \quad (17.5-17) \end{aligned}$$

Integration of Concentration Profile over (Infinite) Film Thickness at $z = L$

According to an over-all mass balance on the film, the total moles of A transferred per unit time across the gas-liquid interface must be the same as the total molar rate of flow of A across the plane $z = L$, which may be calculated by multiplying the volume rate of flow across the plane $z = L$ by the average concentration at that plane:

$$\begin{aligned} \mathcal{W}_A &= \lim_{\delta \rightarrow \infty} (W \delta v_{\max}) \left(\frac{1}{\delta} \int_0^\delta c_A|_{z=L} dx \right) \\ &= W v_{\max} \int_0^\infty c_A|_{z=L} dx \\ &= W v_{\max} c_{A0} \int_0^\infty \text{erfc} \frac{x}{\sqrt{4 \mathcal{D}_{AB} L / v_{\max}}} dx \\ &= W v_{\max} c_{A0} \cdot \frac{2}{\sqrt{\pi}} \int_0^\infty \left(\int_{\frac{x}{\sqrt{4 \mathcal{D}_{AB} L / v_{\max}}}}^\infty e^{-\xi^2} d\xi \right) dx \\ &= W v_{\max} c_{A0} \sqrt{\frac{4 \mathcal{D}_{AB} L}{v_{\max}}} \cdot \frac{2}{\sqrt{\pi}} \cdot \int_0^\infty \left(\int_u^\infty e^{-\xi^2} d\xi \right) du \quad (17.5-18) \end{aligned}$$

In the last line the new variable $u = x/\sqrt{4 \mathcal{D}_{AB} L / v_{\max}}$ has been introduced.

At this juncture we change the order of integration³ in the double integral, which procedure enables us to evaluate the double integral analytically and to obtain Eq. 17.5-17 thus:

$$\begin{aligned} \mathcal{W}_A &= W L c_{A0} \sqrt{\frac{4 \mathcal{D}_{AB} v_{\max}}{\pi L}} \cdot 2 \int_0^\infty e^{-\xi^2} \left(\int_0^\xi du \right) d\xi \\ &= W L c_{A0} \sqrt{\frac{4 \mathcal{D}_{AB} v_{\max}}{\pi L}} \cdot 2 \int_0^\infty e^{-\xi^2} \xi d\xi \\ &= W L c_{A0} \sqrt{\frac{4 \mathcal{D}_{AB} v_{\max}}{\pi L}} \quad (17.5-19) \end{aligned}$$

From this development it is seen that the mass transfer rate is directly proportional to the square root of the diffusivity and inversely proportional to the square root of the "exposure time," $t_{\text{exp}} = L/v_{\max}$. This approach for studying gas absorption was apparently first proposed by Higbie.⁴

Example 17.5-1. Gas Absorption from Rising Bubbles

Estimate the rate at which gas bubbles of A are absorbed by liquid B as the gas bubbles rise at their terminal velocity v_t through a "clean" quiescent liquid.

Solution. For bubbles of moderate size rising in liquids with no surface-active agents, the gas in the bubble undergoes a toroidal circulation as shown in Fig. 17.5-2. As the gas circulates, it encounters fresh liquid at the top of the bubble. As the bubble rises, the liquid moves downward in relation to the bubble and leaves when it reaches the bottom of the bubble. The liquid near the interface is usually in laminar flow and appears to maintain its identity. Thus the liquid behaves much like the liquid at the surface of the falling film just discussed. The contact time is certainly short, so that the penetration of the dissolved gas is slight and the assumptions introduced in §17.5 are valid. Therefore, to a first approximation, we can use the result in Eq. 17.5-17 (or 19) to estimate the rate of gas absorption and the change in bubble size.

The average rate of mass transfer of A is

$$(N_A)_{\text{avg}} = \sqrt{\frac{4 \mathcal{D}_{AB}}{\pi t_{\text{exp}}}} c_{A0} \quad (17.5-20)$$

in which c_{A0} is the solubility of gas A in liquid B . The exposure time t_{exp} is the time required for the liquid to slide along the bubble from top to bottom. This can be taken to be approximately⁵ $t_{\text{exp}} = D/v_t$, in which D is the bubble diameter and

³ See, for example, G. B. Thomas, Jr., *Calculus and Analytic Geometry*, Addison-Wesley Reading, Mass. (1953), Chapter 15.

⁴ R. Higbie, *Trans. A.I.Ch.E.*, **31**, 365-389 (1935).

⁵ R. Higbie, *op. cit.*

v_t is the terminal velocity of the rising bubble. Hence the absorption rate through the bubble-liquid interface is

$$(N_A)_{\text{avg}} = \sqrt{\frac{4\mathcal{D}_{AB}v_t}{\pi D}} c_{A0} \quad (17.5-21)$$

This result has been substantiated⁶ for gas bubbles of about 0.3–0.5 cm in diameter rising through carefully purified water; small amounts of surface-active agents cause a marked decrease in $(N_A)_{\text{avg}}$ because of the formation of a “skin” around

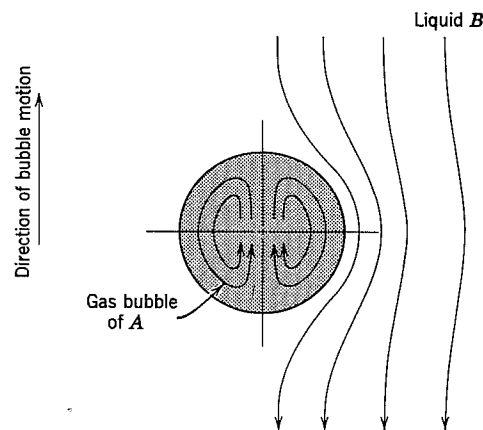


Fig. 17.5-2. Absorption from a gas bubble.

the bubble that effectively prevents circulation. A similar approach has been successfully used for predicting mass transfer rates during drop-formation at a capillary tip.⁷

§17.6 DIFFUSION AND CHEMICAL REACTION INSIDE A POROUS CATALYST: THE “EFFECTIVENESS FACTOR”^{1,2,3}

Up to this point we have discussed diffusion in gases and liquids in systems of simple geometry. We now wish to apply the shell mass-balance method and Fick's first law to describe diffusion on the inside of a porous catalyst pellet. We make no attempt to describe the diffusion inside the tortuous void passages in the medium. Instead, we describe the “average” diffusion of the chemical species in terms of an effective diffusion coefficient.

⁶ D. Hammerton and F. H. Garner, *Trans. Inst. Chem. Engrs. (London)*, **32**, 518 (1954).

⁷ H. Groothuis and H. Kramers, *Chem. Eng. Sci.*, **4**, 17–25 (1955).

¹ E. W. Thiele, *Ind. Eng. Chem.*, **31**, 916–920 (1939).

² R. Aris, *Chem. Eng. Sci.*, **6**, 265–268 (1957).

³ A. Wheeler, *Advances in Catalysis*, Academic Press, New York (1950), Vol. 3, pp. 250–326.

Specifically, we consider a spherical porous catalyst particle of radius R . (See Fig. 17.6-1.) This particle is in a catalytic reactor, where it is submerged in a gas stream containing the reactant A and the product B . In the neighborhood of the surface of the particular catalyst particle under consideration we presume that the concentration is c_{As} moles of A per unit volume. Species A diffuses through the tortuous passages in the catalyst and is converted to B on the catalytic surfaces. (See Fig. 17.6-2.)

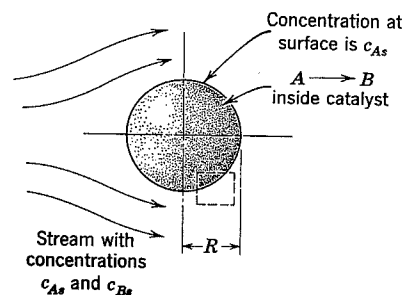


Fig. 17.6-1. Spherical catalyst particle of radius R . For magnified version of inset, see Fig. 17.6-2.

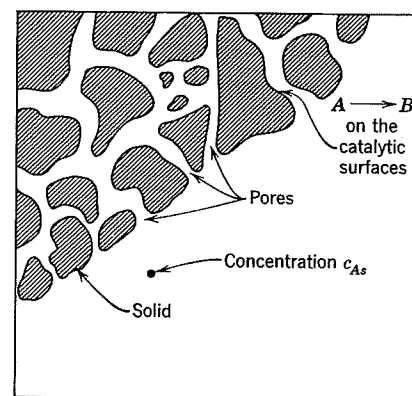


Fig. 17.6-2. Pores in the catalyst in which diffusion and chemical reaction occur.

We begin by making a mass balance for species A on a spherical shell of thickness Δr within a single catalyst particle:

$$N_{Ar}|_r \cdot 4\pi r^2 - N_{Ar}|_{r+\Delta r} \cdot 4\pi(r + \Delta r)^2 + R_A \cdot 4\pi r^2 \Delta r = 0 \quad (17.6-1)$$

Here $N_{Ar}|_r$ is the number of moles of A passing in the r -direction through an imaginary spherical surface at a distance r from the center of the sphere. The source term $R_A \cdot 4\pi r^2 \Delta r$ gives the number of moles of A being produced by chemical reaction in the shell of thickness Δr . Division by $4\pi \Delta r$ and letting $\Delta r \rightarrow 0$ gives

$$\lim_{\Delta r \rightarrow 0} \frac{(r^2 N_{Ar})|_{r+\Delta r} - (r^2 N_{Ar})|_r}{\Delta r} = r^2 R_A \quad (17.6-2)$$

or

$$\frac{d}{dr}(r^2 N_{Ar}) = r^2 R_A \quad (17.6-3)$$

Clearly, in view of the fact that the medium is granular rather than homogeneous, such a limiting process is in conflict with the physical picture. Consequently, in Eq. 17.6-3 the symbols N_{Ar} and R_A cannot be interpreted as quantities having a meaningful value at a point; rather we have to interpret

them as quantities averaged over a small neighborhood of the point in question—a neighborhood small with respect to the dimension R but large with respect to the dimensions of the passages within the porous particle.

Now we define an "effective diffusivity" for species A in the porous medium by

$$N_{Ar} = -\mathcal{D}_A \frac{dc_A}{dr} \quad (17.6-4)$$

in which c_A is the concentration of the gas A contained within the pores. The effective diffusivity $\mathcal{D}_A$ must be measured experimentally; it depends generally on pressure and temperature and also on the catalyst pore structure. The actual mechanism for diffusion in pores is complex, since the pore dimensions may be smaller than the mean free path of the diffusing molecules. We do not belabor the question of mechanism here but assume only that Eq. 17.6-4 can adequately represent the diffusion process.

When the foregoing expression is inserted into Eq. 17.6-3, we get, for constant diffusivity,

$$-\mathcal{D}_A \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dc_A}{dr} \right) = R_A \quad (17.6-5)$$

We now consider a situation wherein species A disappears according to a first-order chemical reaction on the catalytic surfaces that form the "walls" of the winding passages. Let a be the available catalytic surface area per unit volume (of solids + voids). Then $R_A = -k_1''ac_A$ and Eq. 17.6-5 becomes

$$\mathcal{D}_A \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dc_A}{dr} \right) = k_1''ac_A \quad (17.6-6)$$

This equation is to be solved with the boundary conditions that $c_A = c_{As}$ at $r = R$ and that c_A is finite at $r = 0$. Differential equations in which the operator $(1/r^2)(d/dr)[r^2(d/dr)]$ appears can frequently be simplified by a change of variable of the type $c_A/c_{As} = f(r)/r$. The equation for $f(r)$ is

$$\frac{d^2f}{dr^2} = \left(\frac{k_1''a}{\mathcal{D}_A} \right) f \quad (17.6-7)$$

The general solution containing two integration constants is

$$\frac{c_A}{c_{As}} = \frac{C_1}{r} \cosh \sqrt{\frac{k_1''a}{\mathcal{D}_A}} r + \frac{C_2}{r} \sinh \sqrt{\frac{k_1''a}{\mathcal{D}_A}} r \quad (17.6-8)$$

Application of the boundary conditions gives finally

$$\frac{c_A}{c_{As}} = \left(\frac{R}{r} \right) \frac{\sinh \sqrt{k_1''a/\mathcal{D}_A} r}{\sinh \sqrt{k_1''a/\mathcal{D}_A} R} \quad (17.6-9)$$

In studies on chemical kinetics and catalysis one is frequently interested in the molar flux N_{As} or the molar flow $\mathcal{W}_{As}$ at the surface $r = R$:

$$\mathcal{W}_{As} = 4\pi R^2 N_{As} = -4\pi R^2 \mathcal{D}_A \left. \frac{dc_A}{dr} \right|_{r=R} \quad (17.6-10)$$

By making use of Eq. 17.6-9, we find that

$$\mathcal{W}_{As} = 4\pi R \mathcal{D}_A c_{As} \left(1 - \sqrt{\frac{k_1''a}{\mathcal{D}_A}} R \coth \sqrt{\frac{k_1''a}{\mathcal{D}_A}} R \right) \quad (17.6-11)$$

This result gives the rate of conversion (moles/sec) of A to B in a single catalyst particle of radius R in terms of the diffusive processes involved.

If the catalytically active surface were all exposed to the stream of concentration c_{As} , then the species A would not have to diffuse through the pores to a reaction site; the molar rate of conversion would then be given by the product of available surface and surface reaction rate:

$$\mathcal{W}_{A0} = \left(\frac{4}{3}\pi R^3 \right) (a) (-k_1''c_{As}) \quad (17.6-12)$$

Division of Eq. 17.6-11 by Eq. 17.6-12 gives

$$\eta_A = \frac{3}{K^2} (K \coth K - 1) \quad (17.6-13)$$

in which $K = \sqrt{k_1''a/\mathcal{D}_A} R$, a dimensionless group. The quantity η_A is called the *effectiveness factor*.^{1,2,4} It is the quantity by which $\mathcal{W}_{A0}$ has to be multiplied to account for the resistance of diffusion to the over-all conversion process.

For nonspherical catalyst particles, the foregoing results may be applied approximately by reinterpreting R . We note that for a sphere of radius R the ratio of volume to external surface is $R/3$. For nonspherical particles, we redefine R in Eq. 17.6-13 as

$$R_{\text{nonsph}} = 3 \left(\frac{V_P}{S_P} \right) \quad (17.6-14)$$

in which V_P and S_P are the volume and external surface of a single catalyst particle. The conversion rate is then given approximately by

$$|\mathcal{W}_{As}| = V_P a k_1'' c_{As} \eta_A \quad (17.6-15)$$

where

$$\eta_A = \frac{1}{3\Lambda^2} (3\Lambda \coth 3\Lambda - 1) \quad (17.6-16)$$

in which $\Lambda = \sqrt{k_1''a/\mathcal{D}_A} (V_P/S_P)$.

The particular utility of the quantity Λ may be seen in Fig. 17.6-3. It is clear that when the exact theoretical expressions for η_A are plotted as a

⁴ O. A. Hougen and K. M. Watson, *Chemical Process Principles*, Wiley, New York (1947), Part III, Chapter XIX. See also *CPP Charts* by O. A. Hougen, K. M. Watson, and R. A. Ragatz, Wiley, New York (1960), Fig. E.

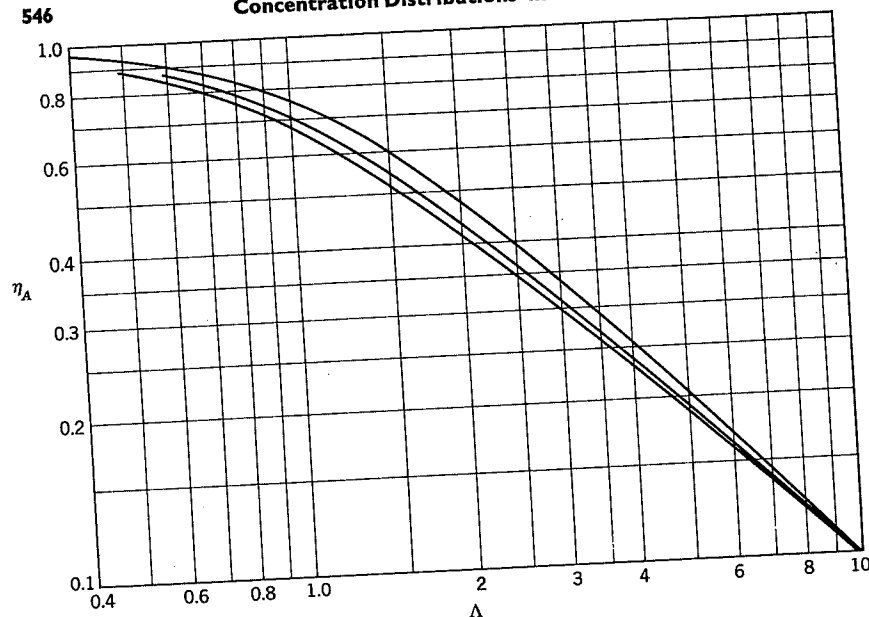


Fig. 17.6-3. Effectiveness factors for porous solid catalysts: top curve, flat particles; middle curve, long cylindrical particles; bottom curve, spherical particles [R. Aris, *Chem. Eng. Sci.*, **6**, 265 (1957).] An enlarged graph for spheres is given in O. A. Hougen, K. M. Watson, and R. A. Ragatz, *CPP Charts*, Wiley, New York (1960), Fig. E.

function of Λ the curves have common asymptotes for large and small Λ and do not differ from one another very much for intermediate values of Λ . Thus Fig. 17.6-3 provides a justification for the use of Eq. 17.6-16 to estimate η_A for nonspherical particles.

QUESTIONS FOR DISCUSSION

1. What physical information is used to eliminate N_{Bz} from Eq. 17.0-1?
2. Compare and contrast *homogeneous* and *heterogeneous* chemical reactions.
3. How would the results of §17.2 be affected if the net motion of the mixture were disregarded [i.e., if $(1 - x_A)$ in Eq. 17.2-4 were set equal to unity]?
4. How do N_{Az} and N_{Bz} vary with distance in §17.2?
5. In Eq. 17.2-9 how is x_{Az} determined?
6. Under what physical situations would you expect the model sketched in Fig. 17.2-1 to be realistic? unrealistic?
7. Discuss the term "diffusion-controlled chemical reaction."
8. Discuss the effect of chemical reaction on gas absorption.
9. Show how Eq. 17.3-14 is obtained from Eq. 17.3-13.

Problems

10. Discuss the physical significance of the approximations made in Eqs. 17.5-4 and 5.
11. Sketch the concentration profiles in Eq. 17.5-15 by looking up a table of the error function.
12. How do the results in §17.5 have to be modified if the film at $z = 0$ contains some dissolved A , say c_{A1} moles per unit volume?
13. Sketch a graph of N_{Az} as a function of z for the problem described by Fig. 17.4-1.
14. How does the size of a soluble bubble change as it moves upward through a liquid?
15. In §17.2 there is a gradient in the concentration of B , yet the flux N_{Bz} is zero. Explain.
16. Use Fig. 17.6-3 to show that for very large Λ the rate of reaction for a given mass of catalyst is proportional to the external surface area of the catalyst particles.
17. How can Eq. 17.3-9 be obtained from Eq. 17.3-2 without getting the concentration profile?

PROBLEMS

17.A₁ Rate of Evaporation

Consider the system described in Fig. 17.2-1. What is the rate of evaporation (in g hr^{-1}) of CCl_3NO_2 (chloropicrin) into air (considered here as a pure substance) at 25°C ?

Total pressure	770 mm Hg
Diffusivity	$0.088 \text{ cm}^2 \text{ sec}^{-1}$
Vapor pressure	23.81 mm Hg
Distance from liquid level to top of tube	11.14 cm
Density of chloropicrin	1.65 g cm^{-3}
Surface area of liquid exposed for evaporation	2.29 cm^2

Answer: 0.0139 g hr^{-1}

17.B₁ Error in Calculating Absorption Rate

What is the maximum possible error in computing the absorption rate from Eq. 17.5-17 if the solubility of A in B is known within ± 5 per cent and the diffusivity of A in B is known within ± 10 per cent? (Assume that the errors in the film dimensions and velocity are known to be of a lower order of magnitude.)

Answer: ± 10 per cent

17.C₁ Rate of Absorption in a Falling Film

Chlorine is being absorbed from a gas in a small experimental wetted wall tower, as shown in Fig. 17.C. The absorbing fluid is water, which is moving with an average velocity of 17.7 cm sec^{-1} . What is the absorption rate in g-moles hr^{-1} if $\mathcal{D}_{\text{Cl}_2-\text{H}_2\text{O}} = 1.26 \times 10^{-5} \text{ cm}^2 \text{ sec}^{-1}$ in the liquid phase and if the saturation concentration of chlorine in water is 0.823 g Cl_2 per 100 g of water (these are the experimental values at 16°C). The dimensions of the column are given in Fig. 17.C.

Hint. Ignore the chemical reaction between Cl_2 and H_2O . Answer: $0.273 \text{ g-moles hr}^{-1}$

17.D₂ Diffusion Method for Separating Helium from Natural Gas

In a short note in *Scientific American*, Vol. 199, No. 1, (July 1958), p. 52, it is described how helium may be separated from natural gas by a method developed by K. B. McAfee of Bell Telephone Laboratories. The method is based on the fact that pyrex glass is almost impermeable to all gases but helium; for example, the diffusion coefficient of He through pyrex is about 25 times the diffusion coefficient of H_2 through pyrex—hydrogen being the closest "competitor" in the diffusion process. This method presumably offers possibility for more efficient and less costly separations than the previous method of low-temperature distillation.

in the somewhat more convenient form:

$$= Sx_{A0} \sqrt{\frac{4\mathcal{D}_{AB}t}{\pi}} \cdot \psi^{(2)} \quad (19.1-21)$$

factor, which accounts for the deviation from Fick's because of the convective contribution (the last ion factor $\psi^{(2)}$ is also given in Table 19.1-1. and for $\mathcal{D}_{AB}$ to give

$$\mathcal{D}_{AB} = \frac{\pi}{t} \left(\frac{V_A}{2Sx_{A0}\psi^{(2)}} \right)^2 \quad (19.1-22)$$

This equation can be used to get diffusivities of volatile liquids.

This problem has been presented here to illustrate that important deviations from Fick's second law of diffusion do occur in evaporating systems.

Example 19.1-2. Unsteady Diffusion with First-Order Reaction^{4,5}

When species A diffuses in a liquid medium B and reacts with it irreversibly ($A + B \rightarrow C$) according to a first-order reaction, then the process of diffusion and homogeneous reaction⁶ is described by

$$\frac{\partial c_A}{\partial t} = \mathcal{D}_{AB} \nabla^2 c_A + k_1''' c_A \quad (19.1-23)$$

provided that the solution of A is dilute and provided that not much C is produced. Here $-k_1'''$ is the rate constant for the homogeneous reaction. Equation 19.1-23 in some cases is to be solved with the initial and boundary conditions:

$$\text{I. C.:} \quad \text{at } t = 0, \quad c_A = 0 \quad (19.1-24)$$

$$\text{B. C.:} \quad \text{at surfaces,} \quad c_A = c_{As} \quad (19.1-25)$$

Here c_{As} may vary with position but must be independent of t . For such problems, show that the solution is⁴

$$c_A = -k_1''' \int_0^t f e^{k_1''' t'} dt' + f e^{k_1''' t} \quad (19.1-26)$$

where f is the solution to Eq. 19.1-23 with $k_1''' = 0$, which also satisfies the boundary and initial conditions in Eqs. 19.1-24 and 25.

Solution. First we test to see whether Eq. 19.1-26 satisfies the differential equation (Eq. 19.1-23); from Eq. 19.1-26 we get the derivatives

$$\frac{\partial c_A}{\partial t} = -k_1''' f e^{k_1''' t} + \frac{\partial f}{\partial t} e^{k_1''' t} + k_1''' f e^{k_1''' t} \quad (19.1-27)$$

$$\nabla^2 c_A = -k_1''' \int_0^t (\nabla^2 f) e^{k_1''' t'} dt' + (\nabla^2 f) e^{k_1''' t} \quad (19.1-28)$$

⁴ P. V. Danckwerts, *Trans. Faraday Soc.*, **47**, 1014-1023 (1951); see also Problem 19.J.

⁵ J. Crank, *The Mathematics of Diffusion*, Oxford University Press (1956), pp. 124-125.

⁶ For some cases neutron diffusion is of this form.

Substitution of Eqs. 19.1-26, 27, and 28 into Eq. 19.1-23 and making use of the fact that $\mathcal{D}_{AB} \nabla^2 f = \partial f / \partial t$ shows that Eq. 19.1-23 is indeed satisfied.

The initial condition is satisfied by Eq. 19.1-26, since

$$c_A(t=0) = -k_1''' \int_0^0 f e^{k_1''' t'} dt' + f(t=0) e^{k_1''' \cdot 0} = 0 \quad (19.1-29)$$

because $f(t=0)$ has been defined to be zero.

The boundary condition is also satisfied, since on the surface $f = c_{As}$:

$$\begin{aligned} c_A(\text{surface}) &= -k_1''' \int_0^t c_{As} e^{k_1''' t'} dt' + c_{As} e^{k_1''' t} \\ &= -k_1''' c_{As} \frac{e^{k_1''' t}}{k_1'''} \Big|_0^t + c_{As} e^{k_1''' t} \\ &= c_{As} \end{aligned} \quad (19.1-30)$$

Example 19.1-3. Gas Absorption with Rapid Chemical Reaction⁷

Gaseous A is absorbed by solvent S , the latter containing solute B . The gas-liquid interface is taken to be the plane $z = 0$. Species A reacts with B according to the reaction $A + B \rightarrow AB$ in an instantaneous irreversible reaction. It may be assumed that Fick's second law adequately describes the diffusion processes, since A , B , and AB are present in S in low concentrations. Obtain expressions for the concentration profiles.

Solution. Because of the instantaneous reaction of A and B , there will be a plane parallel to the liquid-vapor interface at a distance z' from it, which separates the region containing no A from that containing no B . The distance z' is a function of t , since the boundary between A and B retreats as B is used up in the chemical reaction.

The differential equations to be solved are

$$\frac{\partial c_A}{\partial t} = \mathcal{D}_{AS} \frac{\partial^2 c_A}{\partial z^2} \quad 0 < z \leq z'(t) \quad (19.1-31)$$

$$\frac{\partial c_B}{\partial t} = \mathcal{D}_{BS} \frac{\partial^2 c_B}{\partial z^2} \quad z'(t) \leq z < \infty \quad (19.1-32)$$

We now try solutions of the form

$$\frac{c_A}{c_{A0}} = a_1 + a_2 \operatorname{erf} \frac{z}{\sqrt{4\mathcal{D}_{AS}t}} \quad (19.1-33)$$

$$\frac{c_B}{c_{B0}} = b_1 + b_2 \operatorname{erf} \frac{z}{\sqrt{4\mathcal{D}_{BS}t}} \quad (19.1-34)$$

in which c_{A0} is the interfacial liquid-phase concentration of A and c_{B0} is the original

⁷ T. K. Sherwood and R. L. Pigford, *Absorption and Extraction*, McGraw-Hill, New York (1952), pp. 332-337. Other problems of diffusion with chemical reaction are discussed in Chapter IX of this reference. See H. S. Carslaw and J. C. Jaeger, *Heat Conduction in Solids*, Oxford University Press, Second Edition (1959), for heat-conduction problems with moving boundaries associated with phase changes.